

Supplementary Material for “FraudShiftBench: Deployment-Contract Evaluation for Temporal Graph Fraud Detection”

Anonymous Author

Abstract

This supplement is the human-readable technical evidence layer for FraudShiftBench. Its twenty-part reviewer path covers notation, support properties, dataset/model/protocol cards, optimization, matched statistics, aggregate results, resource cases, all 22 typed claim scopes, and reproduction. Four exhaustive row tables—180 node-protocol seeds, 198 legacy paired tests, 840 IBM seeds, and 208 IBM context sensitivities—are retained with schemas and checksums in the machine-readable artifact rather than typeset microscopically. Two construct boundaries remain explicit: IBM early-to-late uses first-50% classifier labels with a shared first-60% label-free history map, and the V24 temporal-stress labels are duplicate metadata on 120 scientific cells. All values come from the frozen saved-evidence baseline; no model is retrained and no blocked cell is assigned performance.

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I. SCOPE AND NAVIGATION

A. What belongs in each review surface

The main paper states the decisive scientific comparisons. This supplement gives the definitions, aggregate evidence, uncertainty, feasibility conditions, and interpretation needed to audit those comparisons. The machine-readable artifact retains exhaustive seed rows, context rows, result paths, manifests, and checksums. This division changes the interface, not the evidence: every value and status is read from the frozen rebuild surfaces.

The admissible evidence families are (i) the Elliptic and DGraphFin protocol grid; (ii) the IBM AML-Data baseline, stronger-graph, and construction studies; and (iii) the saved-prediction GraphSafe case. Internal version names identify locks only. A file is not evidence merely because it contains metrics. This distinction excludes failed imports, duplicate metadata labels, compatibility aliases, and zero-output run plans from empirical comparisons.

TABLE I: Reviewer path through the curated supplement.

Part	Section	Question answered
1	Scope and navigation	Reading order, evidence boundary, and destination map.
2	Notation and formal definitions	Typed tasks, contracts, evidence units, and claims.
3	Support-relation properties and controlled validation	Five properties and the 14 controlled validator cases.
4	Dataset cards	Prediction units, labels, graph structure, counts, and limitations.
5	Temporal windows and visibility matrices	Independent access coordinates, including IBM 50%/60% visibility.
6	Graph-construction definitions	Reference graph and matched structural/feature interventions.
7	Model cards and equations	Feature, node-GNN, edge-classifier, and GINE computations.
8	Optimization, selection, and thresholding	Optimizer, schedule, validation selection, and decision thresholds.
9	Metrics and statistical procedure	Metric estimands, seed blocks, intervals, tests, and correction families.
10	Elliptic/DGraphFin protocol results	Strict-to-isolated paired effects with uncertainty.
11	Robustness and negative controls	V24 construct audit and V22 aggregate robustness evidence.
12	IBM AML baseline results	Eight exact baseline contexts without cross-task pooling.
13	IBM graph construction and scale results	Dependence-aware effects, scale feasibility, and runtime.
14	Rank-versus-decision analyses	Winner and rank disagreement under AUPRC, AUROC, and F1@0.5.
15	Review-budget and calibration diagnostics	Capacity-conditioned review and descriptive calibration.
16	GraphSafe-TTA case study	Bounded saved-prediction policy with positive and negative comparators.
17	Resource-boundary case studies	Five resource cases; no blocked cell receives a predictive value.
18	Claim ledger and provenance map	Five thematic tables preserving all 22 typed claims.
19	Reproduction and artifact index	No-training rebuild, schemas, paths, checksums, and raw-row allocation.
20	Extended limitations and ethics	Construct, statistical, domain, compute, fairness, and intended-use limits.

Scope and reading. The sequence moves from definitions to evidence and then to provenance. Exhaustive rows are not part of this reading path; Table efab:s-raw-artifacts locates them in the artifact.

Table I is also an allocation map. Parts I–IX define the estimands and contracts; Parts X–XVII present aggregate evidence and resource outcomes; Parts XVIII–XX connect claims to provenance, reproduction, limitations, and ethics. A reviewer can therefore follow the technical argument without reading a row-level dump.

B. Frozen evidence boundary

The underlying scope remains 180 validated node-protocol result rows and prediction references, five passing V22 full-10 lanes with 640 results and prediction references, 120 scientifically distinct V24 rerun cells after removing 240 metadata-label duplicates, 840 IBM result/prediction pairs, and the saved GraphSafe analysis aggregated to ten seed blocks. The typed ledger contains 22 claims. Resource-blocked cells remain present but nonnumeric.

Blocked means that a required predicate cannot be evaluated; resource-blocked records a feasibility outcome under a declared envelope; diagnostic-only bars promotion to a primary comparison; excluded records fail integrity or construct requirements;

TABLE II
NOTATION USED BY THE DEPLOYMENT-CONTRACT AND STATISTICAL ANALYSES.

Symbol	Role	Definition
i	Prediction unit	Transaction node, account node, or transaction edge, as declared by the dataset card.
m_i	Eligibility	$\mathbb{K}[y_i \neq 0]$ for node tasks; unknown units never enter supervised masks.
z_i	Binary target	$\mathbb{K}[y_i = 1]$ on eligible node units; IBM uses $z_i \in \{0, 1\}$ directly.
q_i	Saved score	Fraud score on the evaluation set under one exact contract.
Π	Deployment contract	$(\mathcal{T}, \mathcal{V}, \mathcal{C}, \mathcal{S}, \mathcal{B}, \mathcal{R})$.
e	Evidence unit	Dataset, prediction unit, contract, construction, model, seed set, metrics, predictions, resources, and provenance.
c	Typed claim	Exact scope, quantifier, comparison, metric, direction, uncertainty rule, and deployment interpretation.
$\mathcal{E} \models c$	Support	Every required unit is admissible and complete, and the scoped predicate is true.
Δ_s	Paired effect	Candidate-minus-reference or isolated-minus-strict difference for matched seed block s .
d_z	Paired effect size	$\overline{\Delta}/s_{\Delta}$; undefined variance is not converted to evidence.

Scope and reading. The prediction unit is part of the evidence type. Consequently, node, account, and transaction-edge values are never pooled into a single leaderboard.

and refuted-in-scope records complete evidence that contradicts the stated direction. None is a synonym for a low score. This separation addresses familiar leakage and selection concerns without pretending that bookkeeping alone establishes construct validity [1]–[4].

a) Interpretive limit. This supplement documents deterministic reconstruction from validated saved outputs. It does not report a clean-room training rerun, a new dataset, a new model family, an external validator study, or a resolution of venue policy.

II. NOTATION AND FORMAL DEFINITIONS

A. Prediction units and supervised labels

Let i index the declared prediction unit: an Elliptic transaction node, a DGraphFin account node, or an IBM transaction edge. For the node tasks, the repository label code is $y_i \in \{0, 1, 2\}$, where 0 is unknown, 1 is illicit/fraud, and 2 is licit/normal. Supervised eligibility and the binary target are

$$m_i = \mathbb{K}[y_i \neq 0], \quad z_i = \mathbb{K}[y_i = 1] \quad \text{for } m_i = 1. \quad (1)$$

Unknown units may remain as unlabeled graph context but never enter train, validation, or test masks. IBM stores $z_i \in \{0, 1\}$ directly for each materialized transaction. This typing prevents the raw unknown code from turning the supervised task into a false three-class estimand.

B. Contract, evidence unit, and typed claim

Definition 1 (Deployment contract). *A contract is*

$$\Pi = (\mathcal{T}, \mathcal{V}, \mathcal{C}, \mathcal{S}, \mathcal{B}, \mathcal{R}), \quad (2)$$

where the coordinates record time, visibility, construction, preprocessing/selection, decision budget or cost, and resources/status. They are separately recorded and non-substitutable, not statistically independent.

Definition 2 (Evidence unit). *An evidence unit is*

$$e = (D, V, U, \Pi, C, M, H, S, Q, P, R, Z), \quad (3)$$

where U is the prediction unit, H the fixed configuration, S the observed seed set, Q the metric record, P the prediction manifest, R the measured or blocked resource record, and Z the provenance/integrity state.

Definition 3 (Typed claim). *A claim is*

$$c = (\Omega, q, \Delta, r, d, u, \iota), \quad (4)$$

where Ω is exact scope, q the quantifier, Δ the comparison, r the metric, d the direction, u the uncertainty requirement, and ι the deployment interpretation.

Definition 4 (Support). *For evidence collection $\mathcal{E}$, $\mathcal{E} \models c$ only if every required scientific cell and seed passes its canonical lock, prediction-dependent claims have complete prediction references, pairing and uncertainty requirements are met, blocked cells remain nonnumeric, and the scoped predicate is true.*

The notation table is intentionally compact; exact evidence-unit fields and matched identifiers are retained in `results/tkde_rebuild/CLAIM_EVIDENCE_LEDGER.csv`. The formal flow is visualized in Fig. 1.

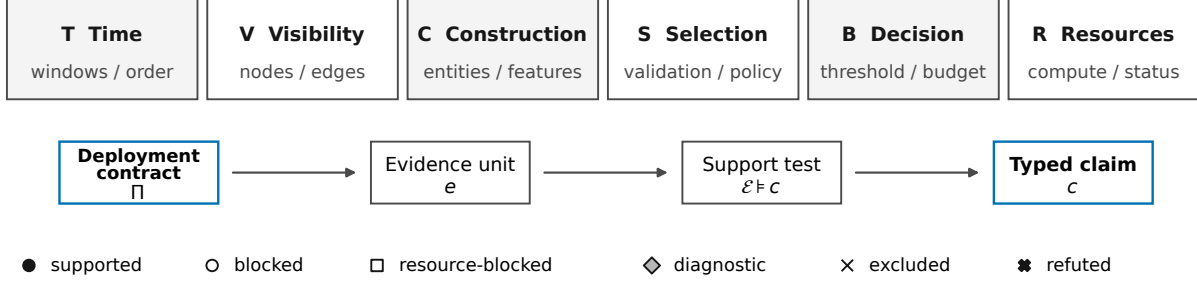


Fig. 1. A deployment contract identifies the evidence unit and the support test applied to a typed claim. Status is an evidence outcome, not a color-coded performance rank.

TABLE III
PROPERTIES ENFORCED BY THE TYPED SUPPORT RELATION.

ID	Property	Operational consequence
P1	Scope monotonicity	Widening a claim adds requirements; it cannot erase requirements already named.
P2	Evidence restriction	Removing the only admissible witness for a required element breaks support.
P3	Rank invariance	Strictly increasing score calibration preserves AUROC/AUPRC ordering but may change a fixed-threshold decision.
P4	Protocol non-substitutability	A result under one claim-relevant visibility or selection coordinate cannot fill another without an invariance argument.
P5	Resource separation	A resource-blocked cell is measurable for feasibility but unordered by predictive performance.

Scope and reading. These are specification and ordering properties, not claims that the ontology is complete or that the validator establishes scientific truth.

a) *Interpretive limit.*: Numerical equality across two contracts does not establish that their information access or estimands are interchangeable. Conversely, a numerical difference does not by itself identify which contract coordinate caused the change.

III. SUPPORT-RELATION PROPERTIES AND CONTROLLED VALIDATION

A. Properties

The support relation treats benchmark reporting as a measurement problem: claims, operational interpretations, and evidence scopes are recorded together [5]–[7]. Table III summarizes the five properties used by the validator.

Proposition 1 (Scope monotonicity of requirements). *If c' widens c by adding datasets, protocols, models, seeds, or resource regimes while holding the predicate fixed, then $\text{Req}(c) \subseteq \text{Req}(c')$.*

Proof sketch. The requirement builder retains every named cell and auxiliary record and adds the widened scope. It never deletes an existing requirement. The property concerns requirements; it does not make truth monotone. $\square$

Proposition 2 (Evidence restriction). *If e is the only admissible witness for an element of $\text{Req}(c)$, then removing e breaks support for c .*

Proof. The required element becomes uncovered. Duplicate files with the same validated scientific key do not create a second independent witness. $\square$

Proposition 3 (Rank invariance under increasing calibration). *If g is strictly increasing and tie order is unchanged, replacing q_i with $g(q_i)$ preserves AUROC and AUPRC but may change decisions at a fixed numerical threshold.*

Proof. The rank metrics depend on score order, which g preserves. The selected set $\{i : q_i \geq \tau\}$ need not equal $\{i : g(q_i) \geq \tau\}$. $\square$

Proposition 4 (Protocol non-substitutability). *A result under Π_1 cannot satisfy a requirement scoped to Π_2 when the contracts differ on a claim-relevant coordinate, absent a separate invariance argument.*

Proof sketch. The contract is part of the scientific key. Changing a claim-relevant coordinate changes admissible information or the evaluation condition. $\square$

Proposition 5 (Resource separation). *A resource-blocked cell has no predictive ordering relation to a measured cell.*

Proof. The blocked cell has no observed AUPRC, AUROC, or F1. It can be compared only as a feasibility outcome under $\mathcal{R}$. $\square$

B. Controlled implementation validation

Table IV reports the complete controlled suite. Each case starts from a frozen claim/evidence relation and changes one required property in a sandbox: scope, seed completeness, prediction presence, integrity, construct identity, resource state, or direction.

TABLE IV: Controlled support-relation validation cases (14 of 14 matched the expected status).

Case	Base claim	Controlled mutation	Expected	Observed
FV01	C01 Elliptic visibility ranking	none; complete locked grid	SUPPORTED	SUPPORTED
FV02	C01 Elliptic visibility ranking	remove one model/protocol cell in sandbox	BLOCKED_ INCOMPLETE_SCOPE	BLOCKED_ INCOMPLETE_SCOPE
FV03	C01 Elliptic visibility ranking	remove seed 10 from one cell in sandbox	BLOCKED_ INCOMPLETE_SEEDS	BLOCKED_ INCOMPLETE_SEEDS
FV04	C01 Elliptic visibility ranking	erase one prediction-manifest reference in sandbox	BLOCKED_MISSING_ PREDICTIONS	BLOCKED_MISSING_ PREDICTIONS
FV05	C10 Small GINE AUPRC result	widen scope from Small to Medium	RESOURCE_BLOCKED	RESOURCE_BLOCKED
FV06	C05 IBM AML baseline grid	widen scope from Small/Medium to all official variants	RESOURCE_BLOCKED	RESOURCE_BLOCKED
FV07	V22 loss robustness	substitute a partial failed import for a full10 lane	EXCLUDED_ INTEGRITY	EXCLUDED_ INTEGRITY
FV08	V24 temporal stress	treat three metadata labels as distinct temporal windows	EXCLUDED_ CONSTRUCT_ INVALID	EXCLUDED_ CONSTRUCT_ INVALID
FV09	DGraphFin fixed GAT performance	promote the T4-OOM cell to a predictive comparison	RESOURCE_BLOCKED	RESOURCE_BLOCKED
FV10	DGraphFin GraphSAGE max-pool rerun	promote runbook/status files without imported rows	RESOURCE_BLOCKED	RESOURCE_BLOCKED
FV11	IBM sender-receiver construction	count the contract alias as an independent replication	EXCLUDED_ CONSTRUCT_ INVALID	EXCLUDED_ CONSTRUCT_ INVALID
FV12	C04 universal graph harm	widen mixed scoped results into a universal directional claim	REFUTED_IN_SCOPE	REFUTED_IN_SCOPE
FV13	C17 universal GraphSafe improvement	widen bounded mixed case study to universal dominance	REFUTED_IN_SCOPE	REFUTED_IN_SCOPE
FV14	IBM early-to-late visibility	describe the shared first-60% label-free node-history map as first-50% training-only features	EXCLUDED_ CONSTRUCT_ INVALID	EXCLUDED_ CONSTRUCT_ INVALID

Scope and reading. The cases localize incomplete scope, missing seeds or predictions, integrity and construct failures, resource blocks, and directional refutation. They are project-authored conformance tests, not external validation.

All 14 observed statuses equal their expected statuses. FV08 rejects the V24 stress-label interpretation because the label never reached the harness; FV11 rejects the sender-receiver alias as an independent method; FV14 rejects a first-50%-only description of IBM account histories. These cases show that the implementation preserves the declared status distinctions.

a) *What this does not establish.*: The schema, cases, and implementation were authored in the same project. Passing the suite does not prove ontology completeness, scientific truth, causal validity, curator agreement, fairness, or responsible deployment.

IV. DATASET CARDS

The three dataset families differ in prediction unit, label eligibility, graph semantics, temporal construction, prevalence, and empirical scope. They are therefore presented as separate cards rather than compressed into one cross-task leaderboard.

A. Elliptic

Elliptic is a real, pseudonymized Bitcoin transaction graph [8]. Nodes are transactions with 165 features and 49 discrete time steps. The loader symmetrizes the 234,355 source edges to 468,710 message-passing arcs. Unknown transactions can contribute unlabeled context but are excluded from supervised denominators.

The card answers which units are scored and how the chronological masks are formed. It does not make the 49 bins equivalent to calendar time or establish behavior on another blockchain.

B. DGraphFin

DGraphFin is a real, anonymized financial-user graph [9]. The task scores account nodes. Raw background classes 2 and 3 are mapped to unknown and excluded from supervised masks. Node time is the median timestamp of incident edges, followed by 20 equal-count buckets; isolated nodes receive the global median timestamp.

The equal-count bucket construction is an implementation choice. It supports the reported fixed split but not a claim about a particular calendar horizon or bank workflow.

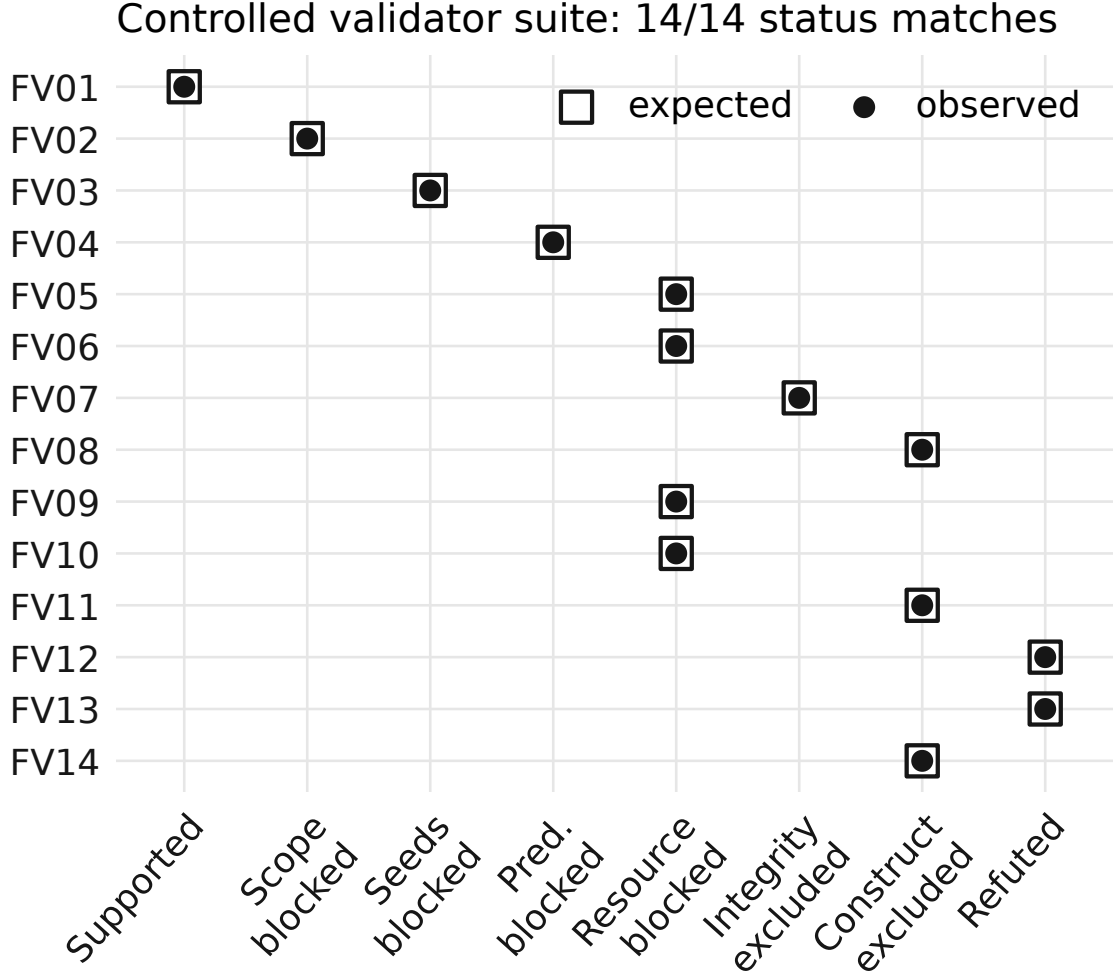


Fig. 2. Observed transitions for the 14 controlled claim/evidence mutations. The suite tests conformance to the project specification, not universal validator correctness.

TABLE V
ELLIPTIC DATASET CARD.

Field	Verified value
Origin	real pseudonymized Bitcoin transactions
Prediction unit	transaction node
Graph size	203,769 nodes; 234,355 source edges; 468,710 message-passing arcs
Features	165 node; 0 edge
Temporal order	49 discrete time steps
Protocols/windows	strict/isolated/transductive; train t1-30, val t31-34, test t35-49
Train/validation/test	26,905 / 2,989 / 16,670 eligible units
Positive counts	2,954 / 508 / 1,083
Test prevalence	6.497%
Label mapping	raw 1=illicit- ζ positive 1; raw 2=licit- ζ supervised 0 (repository code 2); unknown excluded
Source surface	data/raw/elliptic_txs_*.csv; results/runs_rb09v3/runs.csv
Known limitation	The 49 steps are dataset time bins; conclusions do not imply calendar-time or institution-level generalization.

Scope and reading. Counts and label mappings are recovered from the frozen dataset/task statistics. Unknown node labels remain graph context but never enter supervised masks.

TABLE VI
DGRAPHFIN DATASET CARD.

Field	Verified value
Origin	real anonymized financial user graph
Prediction unit	account/user node
Graph size	3,700,550 nodes; 4,300,999 source edges; 7,994,520 message-passing arcs
Features	17 node; 0 edge
Temporal order	20 equal-count incident-edge timestamp buckets
Protocols/windows	strict/isolated/transductive; train t1-14, val t15-16, test t17-20
Train/validation/test	947,462 / 107,932 / 170,207 eligible units
Positive counts	12,543 / 1,103 / 1,863
Test prevalence	1.095%
Label mapping	raw 1=fraud-;positive 1; raw 0=normal-;supervised 0 (repository code 2); raw 2/3 background excluded
Source surface	data/raw/dgraphfin.npz; results/runs_rb09v3/runs.csv
Known limitation	Time is a median incident-edge timestamp placed into equal-count buckets, not a calendar deployment schedule.
<i>Scope and reading.</i> Counts and label mappings are recovered from the frozen dataset/task statistics. Unknown node labels remain graph context but never enter supervised masks.	

TABLE VII
IBM AML-DATA DATASET CARD.

Field	Verified value
Origin	Synthetic financial transactions from IBM AML-Data
Prediction unit	Directed transaction edge
Measured variants	HI MEDIUM, HI SMALL, LI MEDIUM, LI SMALL
Graph range	515,088–2,077,023 accounts; 5,078,345–31,898,238 transactions
Features	Eight transaction attributes and eight account-history attributes
Temporal order	Stable chronological transaction sort
Labels	is_laundering=1 positive; 0 negative; no unknown class in the materialized task
Visibility	Classifier labels use 50% or 60% by protocol; both protocols share the first-60% label-free account-history map
Unmeasured scope	HI/LI Large are guard-blocked; HI/LI Medium GINE h64 is T4-OOM
Known limitation	HI/LI are synthetic generator regimes, not observed bank populations or severity categories
<i>Scope and reading.</i> Predictive evidence exists only for the four Small/Medium variants. Large and Medium-GINE cells are retained as resource outcomes, not assigned performance.	

C. IBM AML-Data

IBM AML-Data is synthetic financial-transaction data, not bank production evidence [10]. The task scores directed transaction edges in HI/LI Small/Medium account networks. HI and LI identify generation regimes with different laundering frequencies; they are not severity labels.

The measured graphs range from roughly 5.1 million to 31.9 million transaction edges. Their scale is useful for resource and construction analysis, but one synthetic generator does not establish representativeness across institutions, typologies, jurisdictions, or adversarial adaptation.

a) *Cross-dataset interpretation.*: Elliptic transaction nodes, DGraphFin account nodes, and IBM transaction edges answer different questions. Their raw AUPRC values, prevalences, and feasible models are retained within typed evidence units and never averaged into a universal score.

V. TEMPORAL WINDOWS AND VISIBILITY MATRICES

A. Node-task contracts

Strict-inductive and isolated-inductive use identical temporal masks and training subgraphs. During evaluation, strict-inductive exposes the held-out structure with labels hidden. Isolated-inductive removes every edge with exactly one endpoint in the held-out union, so held-out nodes can communicate among themselves but not across the train/held-out boundary. The transductive comparator exposes the full graph during training and evaluation while withholding validation/test labels from the loss.

Table VIII separates these access choices from selection and decision rules.

The key interpretation is visibility sensitivity under fixed masks, not a causal decomposition of homophily, degree shift, oversmoothing, or temporal mechanisms.

B. Access matrix

Table IX states labeled access, validation use, test-feature access, graph access, historical covariates, selection/threshold, and resource scope independently.

TABLE VIII
PROTOCOL CARDS KEEP TIME, VISIBILITY, SELECTION, AND DECISION COORDINATES SEPARATE.

Protocol	Masks	Graph/covariate visibility	Labels	Selection/decision	Datasets
Strict	chronological	training nodes/edges only during training; held-out structure available at evaluation	test labels hidden	validation F1	Elliptic, DGraphFin
Isolated	same chronological masks	held-out nodes isolated from training-period cross-split edges at evaluation	test labels hidden	validation F1	Elliptic, DGraphFin
Transductive	same masks	full graph structure visible	test labels hidden	validation F1	Elliptic, DGraphFin
Late-window	60/20/20 chronological	shared node-history map uses first 60%, which is this protocol’s training interval	test transaction labels hidden	fixed 0.5 threshold for saved F1	IBM AML-Data
Early-to-late	50/20/30 chronological	classifier labels use first 50%; shared label-free node-history map uses first 60%, including covariates from 50-60%	test transaction labels hidden	fixed 0.5 threshold for saved F1	IBM AML-Data

Scope and reading. Strict and isolated protocols share masks but differ in graph access. IBM early-to-late is explicitly a 50% labeled-training / 60% label-free-history contract.

TABLE IX: Access matrix for the five evaluated deployment contracts.

Contract	Labeled train	Validation	Test features	Graph nodes/edges	Historical covariates	Decision rule	Resources
Strict	Chronological train	Validation F1	Yes	Held-out structure at evaluation	Training-period only	Validation-selected	Recorded envelope run
Isolated	Same train	Validation F1	Yes	Held-out subgraph; no train/held-out cross edges	Training-period only	Validation-selected	Recorded envelope run
Transductive	Same train	Validation F1	Yes	Full graph during training/evaluation; held-out labels hidden	Full visible graph	Validation-selected	Recorded envelope run
IBM late-window	First 60% labels	Next 20%	Yes	Account graph and transaction features	First 60% label-free history	Fixed 0.5 threshold	F1 Small/Medium measured
IBM early-to-late	First 50% labels	Next 20%	Yes	Account graph and transaction features	Shared first 60%; includes 50–60% label-free covariates	Fixed 0.5 threshold	F1 Small/Medium measured

Scope and reading. Test labels are hidden in every evaluation. Access to test features or graph structure is a declared contract coordinate and must not be confused with access to test labels.

Test labels are never exposed. Test features or structure may be visible when the declared deployment contract permits them. Calling all non-label access *leakage* would collapse scientifically different contracts; failing to disclose it would be equally misleading.

C. IBM temporal windows

IBM late-window uses 60/20/20 chronological masks. Early-to-late uses 50/20/30 classifier masks, but both protocols reuse an account-history matrix constructed from the first 60% of transactions. Thus early-to-late receives label-free covariates from 50–60%, half of its nominal validation interval.

TABLE X: Exact IBM temporal-window counts and test prevalence.

Variant	Protocol	Train	Validation	Test	Test positives	Test prevalence
HI MEDIUM	Early-to-late	15,949,119	6,379,647	9,569,472	14,147	0.148%
HI MEDIUM	Late-window	19,138,942	6,379,648	6,379,648	10,642	0.167%
HI SMALL	Early-to-late	2,539,172	1,015,669	1,523,504	2,321	0.152%
HI SMALL	Late-window	3,047,007	1,015,669	1,015,669	1,797	0.177%
LI MEDIUM	Early-to-late	15,625,741	6,250,297	9,375,445	5,392	0.058%
LI MEDIUM	Late-window	18,750,889	6,250,297	6,250,297	3,783	0.061%
LI SMALL	Early-to-late	3,462,024	1,384,810	2,077,215	1,334	0.064%
LI SMALL	Late-window	4,154,429	1,384,810	1,384,810	925	0.067%

Scope and reading. The table reports prediction-unit denominators. It does not pool HI/LI or Small/Medium, and the early-to-late history visibility remains the shared first-60% map described above.

TABLE XI
MATCHED IBM GRAPH-CONSTRUCTION AND FEATURE INTERVENTIONS.

Construction	Operation	Question	Matched scope
NoEdge	zero transaction attributes; structure retained	tests edge-attribute contribution	matched to h64 reference
ShuffledEdge	edge attributes permuted within temporal masks	destroys edge-feature alignment while preserving marginal values	matched to h64 reference
DegreeOnly	endpoint-degree attributes replace transaction features	tests structural summaries without original attributes	matched to h64 reference
DegreeCap	q=.995 cap on training structure	tests hub sensitivity and resource/performance tradeoff	matched to h64 reference
RecentWindow	most recent 50% of training edges	tests recency and reduced computation	matched to h64 reference
Sender-receiver alias	Restates the already materialized sender-to-receiver transaction edges	Contract identity check; no new transformation	Diagnostic only; never counted as an independent method

Scope and reading. Every intervention retains variant, protocol, seed, optimizer schedule, and h64 reference family unless the operation itself changes training structure. The sender-receiver row is an alias, not replication.

The exact denominators and prevalences explain why raw values remain context-specific. The shared 60% history is not test-label leakage, yet it prevents a pure first-50%-only information claim and blocks causal attribution of protocol differences to label-window timing alone.

VI. GRAPH-CONSTRUCTION DEFINITIONS

The IBM construction study uses the edge-aware SAGE-derived h64 classifier as its reference. Every intervention retains variant, protocol, seed, and optimizer schedule unless the intervention itself changes the training structure. Table XI records the operation and question for each matched candidate.

a) Feature controls.: NoEdge sets all eight transaction attributes to zero while preserving account structure and histories. ShuffledEdge permutes transaction-feature rows separately within train, validation, and test masks, preserving marginal distributions and split sizes while breaking transaction-to-feature alignment. DegreeOnly replaces the transaction attributes with source/destination in/out-degree summaries computed from training edges.

b) Structural controls.: DegreeCap removes training edges incident to accounts above the 0.995 active-training-degree quantile, subject to safety guards. RecentWindow uses only the chronologically latest 50% of training edges to construct the training structure, changing both represented history and computation.

c) Identity diagnostic.: The sender-receiver row records that the materialized arrays already encode sender-to-receiver transactions. It performs no independent graph transformation. Its matched zero rows remain useful as an identity check but cannot count as replication or as another method in a winner set.

d) Interpretive limit.: These interventions isolate implementation-level representation and construction choices within the recorded IBM contracts. They do not identify causal effects for arbitrary AML networks, and Medium GINE remains outside the feasible set.

VII. MODEL CARDS AND EQUATIONS

A. Node classifiers

The feature-only MLP does not consume adjacency and therefore serves as a negative control for the graph-only visibility intervention. The GCN update is

$$H^{(\ell+1)} = \sigma\left(\widehat{D}^{-1/2}\widehat{A}\widehat{D}^{-1/2}H^{(\ell)}W^{(\ell)}\right), \quad (5)$$

with self-loops in $\widehat{A}$ [11]. Mean GraphSAGE uses

$$h_v^{(\ell+1)} = \sigma\left(W^{(\ell)}[h_v^{(\ell)} || |N(v)|^{-1} \sum_{u \in N(v)} h_u^{(\ell)}]\right) \quad (6)$$

and therefore depends differently on available neighborhoods [12].

The card makes the architecture control explicit: Elliptic uses h256/l3 while DGraphFin uses h64/l2. The grid studies three mostly static families and does not support a class-wide conclusion about temporal-state, transformer, or heterogeneous GNNs.

TABLE XII
NODE-CLASSIFIER CARDS FOR ELLIPTIC AND DGRAPHFIN.

Model	Computation	Scientific role	Configuration
MLP	node features only	negative control for graph-visibility intervention	Elliptic h256/l3; DGraphFin h64/l2
GCN	normalized neighborhood aggregation	tests spectral-style smoothing under visibility change	Elliptic h256/l3; DGraphFin h64/l2
GraphSAGE	mean neighborhood aggregation	tests inductive aggregation under visibility change	Elliptic h256/l3; DGraphFin h64/l2

Scope and reading. The MLP is the graph-visibility negative control. The grid does not establish class-wide behavior for modern temporal or heterogeneous GNNs.

TABLE XIII
IBM TRANSACTION-CLASSIFIER CARDS.

Model	Input/computation	Scientific role	Configuration
Logistic regression	standardized transaction features	linear non-graph baseline	balanced classes; max_iter=500
Histogram gradient boosting	transaction features	nonlinear tabular baseline	160 iterations; learning rate 0.06
GraphSAGE-derived edge classifier h32	training-history endpoint/neighborhood summaries plus edge features	small IBM graph baseline	20 epochs; batch 65,536
Edge-aware GraphSAGE-derived edge classifier h64	training-history endpoint/neighborhood summaries plus edge features	reference for graph-feature/construction ablations	30 epochs; batch 32,768
GINE h64	one GINE layer and edge head	edge-conditioned message-passing architecture	30 epochs; batch 32,768; Small only

Scope and reading. The h32/h64 systems are static one-hop GraphSAGE-derived edge classifiers. Medium GINE is resource-blocked and has no performance value.

B. IBM transaction classifiers

Logistic regression and histogram gradient boosting operate on transaction attributes without account-graph message passing [13]. The SAGE-derived edge classifiers first form one-hop training-history summaries

$$\bar{x}_{N(v)} = \frac{1}{\max(1, |N_{\text{tr}}(v)|)} \sum_{u \in N_{\text{tr}}(v)} x_u, \quad \tilde{x}_v = [x_v \| \bar{x}_{N(v)}], \quad (7)$$

then score transaction (u, v) with

$$q_{uv} = \sigma(f_{\theta}([\tilde{x}_u \| \tilde{x}_v \| a_{uv}])). \quad (8)$$

They are static one-hop edge classifiers, not stacks of sampled GraphSAGE convolutions.

The feasible GINE candidate uses one edge-conditioned update,

$$h'_v = \text{MLP} \left((1 + \epsilon)h_v + \sum_{u \in N(v)} \text{ReLU}(h_u + \phi(a_{uv})) \right), \quad (9)$$

followed by an edge head [14].

GINE h64 is measured only on Small. The two Medium lanes exhausted a T4 and receive no performance score. The model card preserves that feasibility condition alongside the computational definition.

VIII. OPTIMIZATION, SELECTION, AND THRESHOLDING

Table XIV consolidates optimizer, learning rate, regularization, loss, schedule, validation selection, threshold, and batching. These choices are scientific coordinates because changing selection or threshold can change a decision-level conclusion even when score rankings are fixed.

TABLE XIV: Optimization, validation selection, and decision settings.

Family	Optimizer	LR	Weight decay	Loss	Iterations	Selection	Decision/batch
Elliptic protocol grid	AdamW	1e-3	5e-4	class-weighted CE	200	validation F1	argmax over 3 repository label codes; full graph forward

Continued on next page

TABLE XIV: Optimization, validation selection, and decision settings. (continued)

Family	Optimizer	LR	Weight decay	Loss	Iterations	Selection	Decision/batch
DGraphFin protocol grid	AdamW	1e-3	5e-4	class-weighted CE	200	validation F1	argmax over 3 repository label codes; full graph forward
IBM LR	LBFGS family (scikit-learn)	library default	library default	balanced weight	class max_iter 500	training only	score $\hat{\epsilon} = .5$; CPU
IBM HistGB	histogram gradient boosting	.06	n/a	binary log loss	160	training only	score $\hat{\epsilon} = .5$; CPU
IBM GraphSAGE-derived h32	AdamW	1e-3	1e-4	positive-weighted BCE	20	fixed seed	score $\hat{\epsilon} = .5$; edge minibatches 65,536
IBM h64 graph grid	AdamW	1e-3	1e-4	positive-weighted BCE	30	fixed seed	score $\hat{\epsilon} = .5$; edge minibatches 32,768

Scope and reading. Node-grid checkpoints are selected by validation F1. IBM saved F1 uses 0.5. Deterministic library baselines may repeat exactly across nominal seeds; those repetitions are not reinterpreted as independent stochastic fits.

A. Node-grid selection

Elliptic and DGraphFin use AdamW, class-weighted cross-entropy, cosine learning-rate decay, and a 200-epoch cap. Validation F1 is checked at epoch 1 and every ten epochs. Configured patience is 40 validation checks, so it cannot fire within the cap; the saved row is the best scheduled validation checkpoint. Python, NumPy, and PyTorch seeds are fixed per run, and features are standardized on training rows.

B. IBM fitting and decisions

The IBM tree and logistic baselines run on CPU. The h32 and h64 graph-derived classifiers use positive-weighted binary cross-entropy and edge minibatches. Saved IBM F1 uses threshold 0.5. No later threshold search is retroactively applied to those rows. Runtime records report the actual backend; a legacy path token containing P100 is not hardware evidence and is normalized to the recorded T4/CUDA lane.

C. Deterministic baselines

Some library baselines repeat exactly across nominal seeds. Their ten rows document repeated pipeline evaluation and confirm deterministic output under the saved configuration, but they are not reinterpreted as ten independent stochastic fits. Zero variance remains visible in the aggregate intervals.

a) Interpretive limit. The settings reproduce the saved evidence surface. They do not establish that the hyperparameter grid is exhaustive or optimal for another institution, time period, hardware stack, or model implementation.

IX. METRICS AND STATISTICAL PROCEDURE

A. Estimands

AUPRC is the primary ranking metric under severe imbalance because ROC curves can obscure low precision [15]. AUROC, fixed-threshold F1, review-budget precision/recall, calibration, and illustrative cost answer different questions. Table XV states each estimand and its boundary.

At review fraction b , $K = \lceil b|I| \rceil$ and the top K scores form the queue. The cost scenario is

$$R_c(\tau) = \frac{FP(\tau) + 5FN(\tau)}{|I|}, \quad (10)$$

which is a 1:5 illustration rather than an estimate of bank loss [16]. Calibration is summarized by Brier score and ten-bin ECE; ECE remains binning-dependent [17].

B. Paired uncertainty and correction

Primary cells use seeds 1–10. Arithmetic means and sample standard deviations describe the saved rows. Confidence intervals use 10,000 deterministic percentile-bootstrap resamples with random seed 20260710 [18]. Paired effects match seed s on each side, and

$$d_z = \frac{\bar{\Delta}}{s_{\Delta}}. \quad (11)$$

Two-sided Wilcoxon signed-rank tests operate on nonzero paired differences; all-zero contrasts receive $p = 1$. Holm correction is applied within the declared primary families [19]. The older V22 robustness table retains its existing Benjamini-Hochberg values and is not mixed into the new families [20].

For IBM construction effects, the four fixed HI/LI-by-protocol context differences are averaged within seed before inference, yielding ten seed blocks rather than 40 exchangeable pairs. GraphSafe similarly averages six contexts within seed before its ten-pair analysis. The statistical-family table makes these independent units explicit.

TABLE XV
METRIC CARDS DISTINGUISH RANKING, DECISION, CAPACITY, AND CALIBRATION QUESTIONS.

Metric	Type	Estimand	Interpretation boundary
AUPRC	Rank	Area under precision-recall curve	Primary rare-event ranking metric; interpreted against within-cell prevalence.
AUROC	Rank	Probability a random positive outranks a random negative	Can remain high when precision is poor under severe imbalance.
F1@0.5	Decision	Harmonic mean of precision and recall	IBM operating point only; not threshold-invariant.
Precision/Recall@b	Capacity	Top K scores where K is the declared review fraction	Reported at 0.5%, 1%, and 2%; queue-capacity conditioned.
Normalized AUPRC	Diagnostic	AUPRC divided by test prevalence	Lift context only; not a prevalence-invariant universal metric.
Brier score	Calibration	Mean squared probability error	Descriptive on saved predictions; lower is better.
ECE	Calibration	Ten-bin expected calibration error	Binning dependent and descriptive.
Cost risk	Decision	FP + 5 FN divided by evaluation size	Illustrative 1:5 scenario, not bank monetary loss.

Scope and reading. No metric substitutes for another. In particular, monotone calibration cannot repair an AUPRC ranking, and fixed-threshold F1 cannot stand in for every review policy.

TABLE XVI
STATISTICAL FAMILIES AND INFERENCE UNITS.

Family	Scope	Independent unit	Interval/test	Correction
Node visibility	18 dataset-model-metric effects	10 paired seeds	10,000-resample percentile CI; paired dz; Wilcoxon	Holm within metric
IBM construction	Candidate vs h64 by size and metric	10 seed blocks after averaging four fixed contexts within seed	10,000-resample percentile CI; paired dz; Wilcoxon	Holm within size-by-metric
IBM context sensitivity	208 context-specific effects	10 paired seeds per fixed context	Descriptive CI, dz, and Wilcoxon	Context table is sensitivity, not a second confirmatory family
GraphSafe	48 declared comparator-metric tests	10 seed blocks after averaging six contexts within seed	Percentile CI and Wilcoxon	One Holm family of 48
V22 robustness	198 legacy paired tests	10 paired seeds	Bootstrap CI and paired test	Original Benjamini-Hochberg values retained

Scope and reading. Ten seeds quantify optimization variation on fixed datasets and splits. They do not estimate population variation across banks, institutions, or future periods.

a) *What the intervals do not establish.*: Ten seeds quantify optimization variation on fixed datasets and splits. Resampling does not create institution-level or future-period observations. A confidence interval excluding zero does not override a nonsignificant family-adjusted test, and resource-blocked cells remain absent rather than imputed.

X. ELLIPTIC AND DGRAPHFIN PROTOCOL RESULTS

A. Scientific question and matched scope

The primary question is whether changing only strict-versus-isolated graph visibility leaves each architecture invariant. Both sides use the same dataset, temporal masks, model configuration, and seeds. Tables XVII–XIX report all 18 aggregate effects; the complete 180 seed rows remain in the artifact.

B. AUPRC

The MLP delta is exactly zero on both datasets, as expected for a graph-only intervention. On Elliptic, GraphSAGE changes from 0.61139 to 0.47969 ($\Delta = -0.13170$), while GCN changes from 0.46664 to 0.56046 ($\Delta = 0.09382$); both Holm values are 0.0117. The leading model among MLP, GCN, and GraphSAGE therefore changes from GraphSAGE under strict visibility to GCN under isolated visibility. DGraphFin GraphSAGE declines from 0.02270 to 0.01781 ($\Delta = -0.00489$, Holm $p = 0.0156$), but remains the top model in the tested set.

C. AUROC

AUROC responses do not supply a common sign that could replace the AUPRC result. The aggregate table is retained because rank metrics can disagree, but the paper’s Elliptic leaderboard statement is specifically scoped to AUPRC.

TABLE XVII
STRICT-TO-ISOLATED PAIRED AUPRC EFFECTS.

Dataset	Model	n	Strict	Isolated	Δ [95% CI]	d_z	Holm p
DGraphFin	GCN	10	0.01164	0.01183	+0.00019 [-0.00134, 0.00191]	0.069	1.0000
DGraphFin	MLP	10	0.01398	0.01398	+0.00000 [0.00000, 0.00000]	0.000	1.0000
DGraphFin	GraphSAGE	10	0.02270	0.01781	-0.00489 [-0.00640, -0.00331]	-1.877	0.0156
Elliptic	GCN	10	0.46664	0.56046	+0.09382 [0.07898, 0.10818]	3.765	0.0117
Elliptic	MLP	10	0.45216	0.45216	+0.00000 [0.00000, 0.00000]	0.000	1.0000
Elliptic	GraphSAGE	10	0.61139	0.47969	-0.13170 [-0.14221, -0.12114]	-7.397	0.0117

Scope and reading. All comparisons share seeds 1–10 and identical temporal masks. The intervention changes graph visibility; it does not identify a unique mechanism for the observed response.

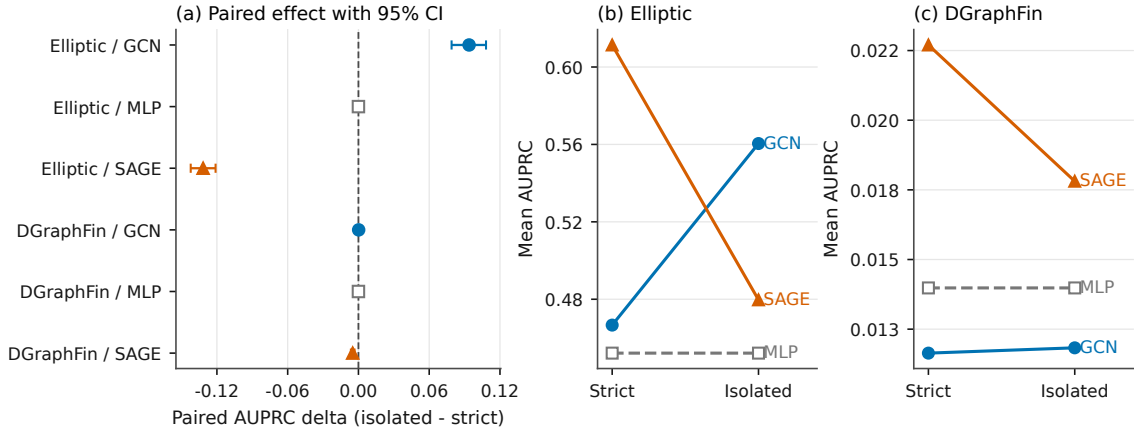


Fig. 3. Paired strict-to-isolated AUPRC effects. The feature-only control remains invariant while graph-model direction and magnitude depend on architecture and dataset.

TABLE XVIII
STRICT-TO-ISOLATED PAIRED AUROC EFFECTS.

Dataset	Model	n	Strict	Isolated	Δ [95% CI]	d_z	Holm p
DGraphFin	GCN	10	0.53049	0.52091	-0.00958 [-0.05269, 0.03227]	-0.131	1.0000
DGraphFin	MLP	10	0.52554	0.52554	+0.00000 [0.00000, 0.00000]	0.000	1.0000
DGraphFin	GraphSAGE	10	0.68688	0.65789	-0.02899 [-0.03921, -0.01956]	-1.748	0.0117
Elliptic	GCN	10	0.87228	0.87002	-0.00226 [-0.00769, 0.00195]	-0.270	1.0000
Elliptic	MLP	10	0.87458	0.87458	+0.00000 [0.00000, 0.00000]	0.000	1.0000
Elliptic	GraphSAGE	10	0.88744	0.86737	-0.02007 [-0.02172, -0.01816]	-6.602	0.0117

Scope and reading. All comparisons share seeds 1–10 and identical temporal masks. The intervention changes graph visibility; it does not identify a unique mechanism for the observed response.

TABLE XIX
STRICT-TO-ISOLATED PAIRED F1 EFFECTS.

Dataset	Model	n	Strict	Isolated	Δ [95% CI]	d_z	Holm p
DGraphFin	GCN	10	0.01642	0.01700	+0.00058 [-0.00266, 0.00386]	0.105	1.0000
DGraphFin	MLP	10	0.02358	0.02358	+0.00000 [0.00000, 0.00000]	0.000	1.0000
DGraphFin	GraphSAGE	10	0.04764	0.03486	-0.01278 [-0.01821, -0.00754]	-1.391	0.0195
Elliptic	GCN	10	0.49932	0.52759	+0.02827 [0.01060, 0.04238]	1.024	0.0781
Elliptic	MLP	10	0.47719	0.47719	+0.00000 [0.00000, 0.00000]	0.000	1.0000
Elliptic	GraphSAGE	10	0.58470	0.53461	-0.05009 [-0.05890, -0.04140]	-3.298	0.0117

Scope and reading. All comparisons share seeds 1–10 and identical temporal masks. The intervention changes graph visibility; it does not identify a unique mechanism for the observed response.

TABLE XX
CONSTRUCT AUDIT OF THE V24 TEMPORAL-STRESS LABELS.

Scientific base cells	Metadata labels/cell	Duplicate rows	Audit finding	Admissible use
120 dataset/protocol/model/seed cells	3	240	The label was not passed to the benchmark harness; every performance metric repeats across labels.	One deduplicated representative per base cell as supplementary rerun evidence only.
<i>Scope and reading.</i> Runtime can differ because the same harness was executed again. No temporal-stress contrast is admissible from this family.				

TABLE XXI
CANONICAL V22 ROBUSTNESS LANES AND FEASIBILITY STATUS.

Scientific lane	Status	Result rows	Prediction files	Prediction rows
DGraphFin loss robustness	PASS_FULLL10_MERGED_ WITH_COMPATIBLE_ ALIASES	180	180	30,637,260
Elliptic loss robustness	PASS_FULLL10_MERGED_ WITH_COMPATIBLE_ ALIASES	180	180	3,000,600
DGraphFin negative controls	PASS_FULLL10	80	80	13,616,560
Elliptic negative controls	PASS_FULLL10	160	160	2,667,200
DGraphFin fixed GAT h64/l2	BLOCKED_T4_OOM	0	0	0
Elliptic GAT/GIN	PASS_FULLL10	40	40	666,800

Scope and reading. Five lanes are complete. DGraphFin GAT h64/l2 is T4-OOM; the memory-reduced diagnostic is a different configuration and cannot fill this cell.

D. F1

F1 is checkpoint/decision-rule dependent and must be interpreted alongside the selection description in Section VIII. Its paired effects confirm that a graph visibility change can alter decision-level behavior, but they do not imply that isolation is uniformly beneficial or harmful.

a) Boundary.: These results show architecture- and dataset-dependent sensitivity to the recorded visibility intervention. They do not identify a causal mechanism such as homophily, degree shift, or oversmoothing, and they do not generalize to every GNN family.

XI. ROBUSTNESS AND NEGATIVE-CONTROL RESULTS

A. V24 construct audit

The V24 runner iterated over three temporal-stress strings, but the string was written only to metadata and was never passed to the benchmark harness. Table XX summarizes the consequence across the complete family.

For each of 120 dataset/protocol/model/seed base cells, all reported performance metrics are identical under the three labels. Runtime can differ because the harness was rerun. The admissible evidence is therefore one deduplicated representative per base cell as a supplementary protocol rerun; no three-stress comparison is supported.

B. V22 lane completeness

The older V22 extension covers loss functions, graph negative controls, and additional Elliptic architectures. Before reading its statistics, Table XXI establishes which lanes are complete under the canonical lock.

Five full-10 lanes contain 640 result and prediction references. The fixed DGraphFin GAT h64/l2 lane is T4-OOM with zero of 20 planned outputs. A memory-reduced h32/l1 diagnostic changes width and depth and therefore cannot fill the blocked cell. Three other raw imports containing 38 result and 38 prediction files fail integrity or comparability checks and remain excluded.

C. V22 statistical-family overview

The complete V22 table contains 198 model-identified paired tests. Table XXII replaces that microscopic row dump with counts by family and metric, direction, and original BH decision.

The aggregate shows the composition of the legacy family without creating a new family or selecting only favorable rows. Exact left/right definitions, intervals, effect sizes, raw p-values, and BH values remain in `manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv`, indexed in Section XIX.

a) What robustness does not establish.: These extensions probe losses, controls, and a limited architecture set. They do not repair the fixed blocked GAT cell, validate the V24 stress labels, or widen the primary graph-learning breadth beyond the measured configurations.

TABLE XXII
AGGREGATE VIEW OF THE 198 V22 PAIRED TESTS.

Family	Metric	Tests	$\Delta > 0$	$\Delta < 0$	BH $p_{0.05}$	Machine-readable source
RB28	AUPRC	36	21	15	18	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB28	AUROC	36	15	21	17	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB28	F1	36	13	23	20	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB29	AUPRC	28	10	18	20	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB29	AUROC	28	11	17	20	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB29	F1	28	14	14	23	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB30	AUPRC	2	2	0	0	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB30	AUROC	2	2	0	1	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv
RB30	F1	2	0	2	2	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv

Scope and reading. Signs are left-minus-right as declared in the source. This aggregate is a navigation aid; it does not create a new hypothesis family or mix the older BH values with the primary Holm families.

TABLE XXIII
IBM BASELINE-GRID AUPRC MEANS AND 95% SEED INTERVALS.

Variant	Protocol	HistGB	LogReg	SAGE-edge h32
HI MEDIUM	Early-to-late	0.08898 [0.08349, 0.09460]	0.01393 [0.01393, 0.01393]	0.02484 [0.02126, 0.02827]
HI MEDIUM	Late-window	0.16928 [0.16298, 0.17578]	0.01572 [0.01572, 0.01572]	0.03712 [0.02973, 0.04393]
HI SMALL	Early-to-late	0.06574 [0.06019, 0.07156]	0.01439 [0.01439, 0.01439]	0.00384 [0.00322, 0.00480]
HI SMALL	Late-window	0.09642 [0.09142, 0.10215]	0.01726 [0.01726, 0.01726]	0.00711 [0.00476, 0.00973]
LI MEDIUM	Early-to-late	0.01425 [0.01351, 0.01515]	0.00370 [0.00370, 0.00370]	0.00693 [0.00539, 0.00820]
LI MEDIUM	Late-window	0.02970 [0.02609, 0.03408]	0.00403 [0.00403, 0.00403]	0.00783 [0.00669, 0.00890]
LI SMALL	Early-to-late	0.01258 [0.01142, 0.01404]	0.00392 [0.00392, 0.00392]	0.00214 [0.00168, 0.00283]
LI SMALL	Late-window	0.01708 [0.01571, 0.01844]	0.00405 [0.00405, 0.00405]	0.00219 [0.00174, 0.00275]

Scope and reading. Each entry is mean [95% bootstrap interval] over ten saved rows. HistGB leads mean AUPRC in these eight exact contexts; the table does not support universal superiority over AML models.

XII. IBM AML BASELINE RESULTS

A. Scientific question and evidence set

The baseline question is: among logistic regression, histogram gradient boosting, and the h32 SAGE-derived edge classifier, how do mean AUPRC values compare across HI/LI, Small/Medium, and the two temporal contracts? The locked V26 surface contains 24 configuration cells, each represented by ten saved rows and prediction exports.

Histogram gradient boosting has the highest mean AUPRC in all eight exact variant/protocol contexts. This is a descriptive result within the three-model synthetic baseline grid. It does not establish that tree models are universally superior for AML, and it does not compare against resource-blocked Large variants.

B. Uncertainty and deterministic rows

Each table entry reports mean and a ten-row bootstrap interval. Some logistic-regression cells are identical across nominal seeds and therefore have zero-width intervals. The table leaves that determinism visible; repeated execution confirms the pipeline output but does not increase independent stochastic evidence.

C. Scale and prevalence

Raw AUPRC is reported because it is the metric used by the claims. HI/LI and Small/Medium differ in test prevalence, so values are not pooled. Normalized AUPRC remains a diagnostic lift over within-cell prevalence rather than a universal, prevalence-invariant score.

a) *Boundary.*: IBM AML-Data is synthetic and the early-to-late covariate map includes the shared first-60% label-free history. Baseline differences cannot be generalized to a production bank or causally attributed to label windows alone.

XIII. IBM GRAPH-CONSTRUCTION AND SCALE RESULTS

A. Dependence-aware matched design

Each candidate is paired to SAGE-edge h64 by variant, protocol, and seed. The four fixed HI/LI-by-protocol differences are averaged within seed before bootstrap and Wilcoxon inference, so $n = 10$, not 40. The following tables report every candidate-size aggregate for AUPRC, AUROC, and F1; all 208 context-specific sensitivities remain machine-readable.

B. AUPRC effects

TABLE XXIV: Dependence-aware matched IBM AUPRC effects relative to SAGE-edge h64.

Size	Candidate	Reference	Candidate	Δ [95% CI]	d_z	Holm p	Context signs
Medium	DegreeCap	0.02844	0.00758	-0.02086 [-0.02248, -0.01935]	-7.854	0.0117	+0 / -4 / 0:0
Medium	DegreeOnly	0.02844	0.01783	-0.01061 [-0.01287, -0.00821]	-2.674	0.0117	+0 / -4 / 0:0
Medium	NoEdge	0.02844	0.01638	-0.01206 [-0.01395, -0.01022]	-3.783	0.0117	+0 / -4 / 0:0
Medium	RecentWindow	0.02844	0.01520	-0.01324 [-0.01543, -0.01092]	-3.408	0.0117	+0 / -4 / 0:0
Medium	Sender-receiver alias	0.02844	0.02844	+0.00000 [0.00000, 0.00000]	0.621	1.0000	+0 / -0 / 0:4
Medium	ShuffledEdge	0.02844	0.01636	-0.01208 [-0.01324, -0.01089]	-6.042	0.0117	+0 / -4 / 0:0
Small	DegreeCap	0.00752	0.01182	+0.00430 [-0.00036, 0.00971]	0.504	0.4648	+3 / -1 / 0:0
Small	DegreeOnly	0.00752	0.00381	-0.00371 [-0.00547, -0.00202]	-1.264	0.0156	+0 / -4 / 0:0
Small	GINE h64	0.00752	0.01942	+0.01190 [0.00934, 0.01447]	2.697	0.0137	+4 / -0 / 0:0
Small	NoEdge	0.00752	0.00485	-0.00267 [-0.00423, -0.00109]	-0.973	0.0820	+0 / -4 / 0:0
Small	RecentWindow	0.00752	0.00418	-0.00334 [-0.00553, -0.00148]	-0.976	0.0137	+0 / -4 / 0:0
Small	Sender-receiver alias	0.00752	0.00752	+0.00000 [0.00000, 0.00000]	0.000	1.0000	+0 / -0 / 0:4
Small	ShuffledEdge	0.00752	0.00422	-0.00330 [-0.00487, -0.00197]	-1.329	0.0137	+0 / -4 / 0:0

Scope and reading. Inference uses ten seed blocks after averaging four fixed HI/LI-by-protocol contexts within seed. The context-sign column summarizes heterogeneity; the complete 208 context rows remain in the artifact. The sender-receiver row is an identity diagnostic.

NoEdge changes mean AUPRC by -0.00267 on Small and -0.01206 on Medium. All four context means are negative at each scale, but only Medium survives Holm correction ($p = 0.0117$, versus 0.0820 on Small). ShuffledEdge changes AUPRC by -0.00330 and -0.01208 , with Holm values 0.0137 and 0.0117 . These matched controls support a bounded role for transaction-feature alignment in this implementation, not a causal statement about all AML graphs.

GINE improves Small AUPRC by 0.01190 (95% CI $[0.00934, 0.01447]$, Holm $p = 0.0137$) with all four context means positive. Medium GINE is absent because it exhausted T4 memory; Small-only performance is not averaged into a complete Small/Medium claim.

C. AUROC effects

TABLE XXV: Dependence-aware matched IBM AUROC effects relative to SAGE-edge h64.

Size	Candidate	Reference	Candidate	Δ [95% CI]	d_z	Holm p	Context signs
Medium	DegreeCap	0.89599	0.87608	-0.01991 [-0.03384, -0.00674]	-0.867	0.0742	+1 / -3 / 0:0
Medium	DegreeOnly	0.89599	0.80754	-0.08845 [-0.09960, -0.07698]	-4.644	0.0117	+0 / -4 / 0:0
Medium	NoEdge	0.89599	0.80501	-0.09098 [-0.09994, -0.08143]	-5.730	0.0117	+0 / -4 / 0:0
Medium	RecentWindow	0.89599	0.92124	+0.02525 [0.01610, 0.03484]	1.576	0.0117	+4 / -0 / 0:0
Medium	Sender-receiver alias	0.89599	0.89599	+0.00000 [0.00000, 0.00000]	0.089	1.0000	+0 / -0 / 0:4
Medium	ShuffledEdge	0.89599	0.79679	-0.09920 [-0.11548, -0.08282]	-3.543	0.0117	+0 / -4 / 0:0
Small	DegreeCap	0.84528	0.86881	+0.02354 [0.00573, 0.04076]	0.788	0.1465	+3 / -1 / 0:0
Small	DegreeOnly	0.84528	0.73127	-0.11401 [-0.13059, -0.09730]	-4.039	0.0137	+0 / -4 / 0:0
Small	GINE h64	0.84528	0.85618	+0.01090 [-0.01596, 0.03856]	0.231	1.0000	+2 / -2 / 0:0
Small	NoEdge	0.84528	0.73569	-0.10959 [-0.12200, -0.09520]	-4.844	0.0137	+0 / -4 / 0:0
Small	RecentWindow	0.84528	0.79306	-0.05222 [-0.06542, -0.03977]	-2.426	0.0137	+0 / -4 / 0:0
Small	Sender-receiver alias	0.84528	0.84528	+0.00000 [0.00000, 0.00000]	0.000	1.0000	+0 / -0 / 0:4
Small	ShuffledEdge	0.84528	0.72068	-0.12460 [-0.13473, -0.11382]	-6.880	0.0137	+0 / -4 / 0:0

Scope and reading. Inference uses ten seed blocks after averaging four fixed HI/LI-by-protocol contexts within seed. The context-sign column summarizes heterogeneity; the complete 208 context rows remain in the artifact. The sender-receiver row is an identity diagnostic.

AUROC sometimes moves in the opposite direction from AUPRC. RecentWindow on Medium lowers AUPRC but raises mean AUROC by 0.02525 . This tradeoff is retained rather than reduced to a winner label.

D. F1 effects

TABLE XXVI: Dependence-aware matched IBM F1 effects relative to SAGE-edge h64.

Size	Candidate	Reference	Candidate	Δ [95% CI]	d_z	Holm p	Context signs
Medium	DegreeCap	0.00809	0.01396	+0.00587 [0.00317, 0.00841]	1.308	0.0352	+4 / -0 / 0:0
Medium	DegreeOnly	0.00809	0.00834	+0.00026 [-0.00292, 0.00282]	0.052	1.0000	+1 / -3 / 0:0
Medium	NoEdge	0.00809	0.00739	-0.00070 [-0.00339, 0.00135]	-0.170	1.0000	+1 / -3 / 0:0

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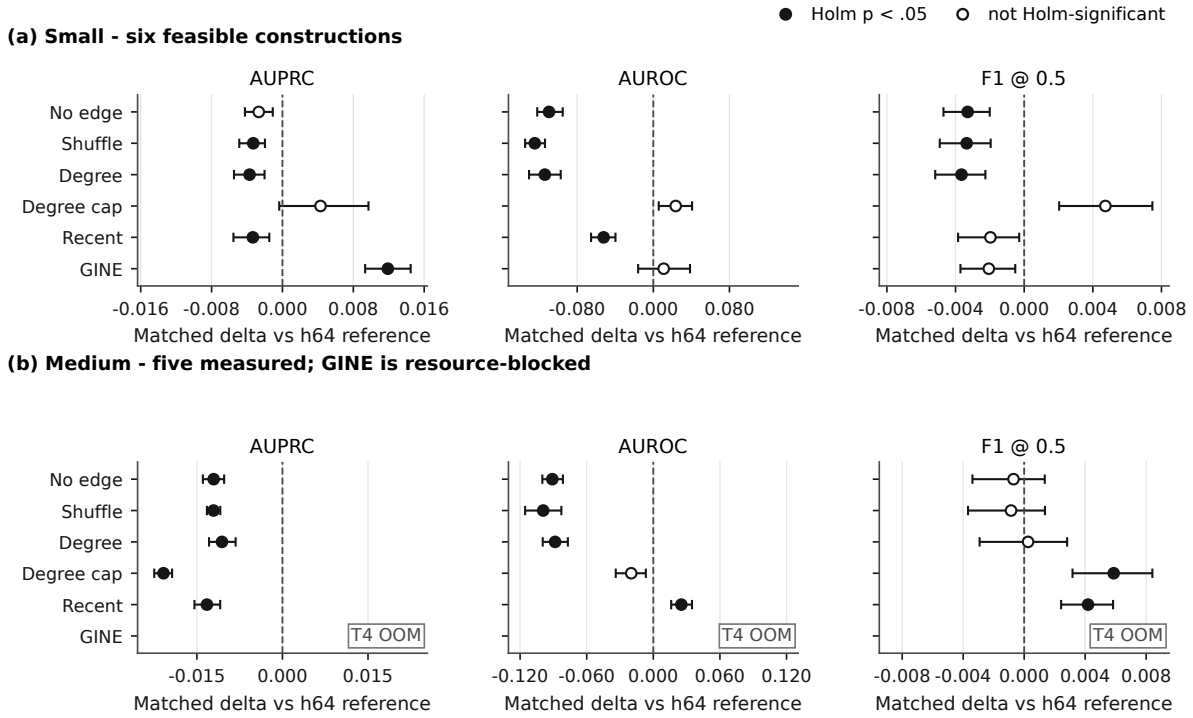


Fig. 4. Matched IBM effects relative to SAGE-edge h64. Intervals use ten seed blocks after fixed-context aggregation; Small-only GINE and blocked Medium cells are not pooled.

TABLE XXVI: Dependence-aware matched IBM F1 effects relative to SAGE-edge h64. (continued)

Size	Candidate	Reference	Candidate	Δ [95% CI]	d_z	Holm p	Context signs
Medium	RecentWindow	0.00809	0.01228	+0.00419 [0.00242, 0.00583]	1.416	0.0352	+3 / -1 / 0:0
Medium	Sender-receiver alias	0.00809	0.00809	+0.00000 [0.00000, 0.00000]	0.000	1.0000	+0 / -0 / 0:4
Medium	ShuffledEdge	0.00809	0.00723	-0.00086 [-0.00368, 0.00136]	-0.199	1.0000	+1 / -3 / 0:0
Small	DegreeCap	0.01058	0.01532	+0.00473 [0.00204, 0.00747]	1.006	0.0781	+4 / -0 / 0:0
Small	DegreeOnly	0.01058	0.00693	-0.00365 [-0.00518, -0.00226]	-1.458	0.0137	+0 / -4 / 0:0
Small	GINE h64	0.01058	0.00852	-0.00206 [-0.00371, -0.00052]	-0.768	0.2520	+0 / -4 / 0:0
Small	NoEdge	0.01058	0.00729	-0.00329 [-0.00470, -0.00201]	-1.452	0.0137	+0 / -4 / 0:0
Small	RecentWindow	0.01058	0.00862	-0.00197 [-0.00385, -0.00029]	-0.645	0.2617	+1 / -3 / 0:0
Small	Sender-receiver alias	0.01058	0.01058	+0.00000 [0.00000, 0.00000]	0.000	1.0000	+0 / -0 / 0:4
Small	ShuffledEdge	0.01058	0.00723	-0.00335 [-0.00492, -0.00195]	-1.327	0.0195	+0 / -4 / 0:0

Scope and reading. Inference uses ten seed blocks after averaging four fixed HI/LI-by-protocol contexts within seed. The context-sign column summarizes heterogeneity; the complete 208 context rows remain in the artifact. The sender-receiver row is an identity diagnostic.

GINE’s Small F1 change is -0.00206 , but its corrected test is not significant ($p = 0.252$). DegreeCap and RecentWindow can improve fixed-threshold F1 while lowering AUPRC. These results are specific to the saved threshold of 0.5.

E. Configuration-specific runtime and feasibility

Table XXVII gives every Small configuration its own AUPRC and elapsed runner time.

TABLE XXVII: IBM Small configuration-level AUPRC, runtime, and feasibility.

Variant	Protocol	Configuration	Status	AUPRC / seconds	Within-cell treatment
HI SMALL	Early-to-late	DegreeCap	MEASURED_FULLL10	0.01285 / 27.89	Pareto
HI SMALL	Early-to-late	DegreeOnly	MEASURED_FULLL10	0.00513 / 31.61	Measured, non-Pareto
HI SMALL	Early-to-late	GINE h64	MEASURED_FULLL10	0.02608 / 264.47	Pareto
HI SMALL	Early-to-late	NoEdge	MEASURED_FULLL10	0.00387 / 35.00	Measured, non-Pareto
HI SMALL	Early-to-late	RecentWindow	MEASURED_FULLL10	0.00496 / 16.64	Pareto
HI SMALL	Early-to-late	SAGE-edge h64	MEASURED_FULLL10	0.01025 / 33.27	Measured, non-Pareto
HI SMALL	Early-to-late	ShuffledEdge	MEASURED_FULLL10	0.00412 / 32.52	Measured, non-Pareto

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TABLE XXVII: IBM Small configuration-level AUPRC, runtime, and feasibility. (continued)

Variant	Protocol	Configuration	Status	AUPRC / seconds	Within-cell treatment
HI SMALL	Late-window	DegreeCap	MEASURED_FULLL10	0.02574 / 39.06	Pareto
HI SMALL	Late-window	DegreeOnly	MEASURED_FULLL10	0.00624 / 44.52	Measured, non-Pareto
HI SMALL	Late-window	GINE h64	MEASURED_FULLL10	0.03125 / 363.32	Pareto
HI SMALL	Late-window	NoEdge	MEASURED_FULLL10	0.01052 / 48.70	Measured, non-Pareto
HI SMALL	Late-window	RecentWindow	MEASURED_FULLL10	0.00682 / 29.69	Pareto
HI SMALL	Late-window	SAGE-edge h64	MEASURED_FULLL10	0.01100 / 47.21	Measured, non-Pareto
HI SMALL	Late-window	ShuffledEdge	MEASURED_FULLL10	0.00752 / 50.12	Measured, non-Pareto
LI SMALL	Early-to-late	DegreeCap	MEASURED_FULLL10	0.00389 / 42.79	Pareto
LI SMALL	Early-to-late	DegreeOnly	MEASURED_FULLL10	0.00168 / 48.76	Measured, non-Pareto
LI SMALL	Early-to-late	GINE h64	MEASURED_FULLL10	0.00974 / 494.76	Pareto
LI SMALL	Early-to-late	NoEdge	MEASURED_FULLL10	0.00217 / 46.50	Measured, non-Pareto
LI SMALL	Early-to-late	RecentWindow	MEASURED_FULLL10	0.00225 / 27.39	Pareto
LI SMALL	Early-to-late	SAGE-edge h64	MEASURED_FULLL10	0.00315 / 50.84	Measured, non-Pareto
LI SMALL	Early-to-late	ShuffledEdge	MEASURED_FULLL10	0.00199 / 46.62	Measured, non-Pareto
LI SMALL	Late-window	DegreeCap	MEASURED_FULLL10	0.00479 / 53.44	Pareto
LI SMALL	Late-window	DegreeOnly	MEASURED_FULLL10	0.00218 / 59.71	Measured, non-Pareto
LI SMALL	Late-window	GINE h64	MEASURED_FULLL10	0.01060 / 675.32	Pareto
LI SMALL	Late-window	NoEdge	MEASURED_FULLL10	0.00283 / 58.44	Measured, non-Pareto
LI SMALL	Late-window	RecentWindow	MEASURED_FULLL10	0.00268 / 35.34	Pareto
LI SMALL	Late-window	SAGE-edge h64	MEASURED_FULLL10	0.00568 / 65.92	Pareto
LI SMALL	Late-window	ShuffledEdge	MEASURED_FULLL10	0.00325 / 58.06	Measured, non-Pareto

Scope and reading. Pareto treatment is computed only within the same variant and protocol. Runtime is runner elapsed time, not a hardware-normalized complexity or energy measure. Blocked rows remain nonnumeric.

GINE takes approximately 9.1 times the h64 reference runtime in the aggregated Small comparison. Its higher AUPRC therefore does not imply universal deployability or dominance under every resource objective.

Table XXVIII gives the corresponding Medium cells and retains blocked GINE rows as nonnumeric statuses.

TABLE XXVIII: IBM Medium configuration-level AUPRC, runtime, and feasibility.

Variant	Protocol	Configuration	Status	AUPRC / seconds	Within-cell treatment
HI MEDIUM	Both Planned Protocols	GINE h64	RESOURCE_BLOCKED_T4_CUDA_OOM	–	Unmeasured
HI MEDIUM	Early-to-late	DegreeCap	MEASURED_FULLL10	0.01186 / 216.79	Measured, non-Pareto
HI MEDIUM	Early-to-late	DegreeOnly	MEASURED_FULLL10	0.02524 / 238.96	Pareto
HI MEDIUM	Early-to-late	NoEdge	MEASURED_FULLL10	0.02447 / 248.13	Measured, non-Pareto
HI MEDIUM	Early-to-late	RecentWindow	MEASURED_FULLL10	0.01866 / 128.15	Pareto
HI MEDIUM	Early-to-late	SAGE-edge h64	MEASURED_FULLL10	0.04242 / 249.73	Pareto
HI MEDIUM	Early-to-late	ShuffledEdge	MEASURED_FULLL10	0.02388 / 255.57	Measured, non-Pareto
HI MEDIUM	Late-window	DegreeCap	MEASURED_FULLL10	0.01274 / 261.86	Measured, non-Pareto
HI MEDIUM	Late-window	DegreeOnly	MEASURED_FULLL10	0.03262 / 290.36	Pareto
HI MEDIUM	Late-window	NoEdge	MEASURED_FULLL10	0.02989 / 298.87	Measured, non-Pareto
HI MEDIUM	Late-window	RecentWindow	MEASURED_FULLL10	0.02895 / 153.10	Pareto
HI MEDIUM	Late-window	SAGE-edge h64	MEASURED_FULLL10	0.04998 / 301.12	Pareto
HI MEDIUM	Late-window	ShuffledEdge	MEASURED_FULLL10	0.02970 / 311.00	Measured, non-Pareto
LI MEDIUM	Both Planned Protocols	GINE h64	RESOURCE_BLOCKED_T4_CUDA_OOM	–	Unmeasured
LI MEDIUM	Early-to-late	DegreeCap	MEASURED_FULLL10	0.00316 / 211.08	Measured, non-Pareto
LI MEDIUM	Early-to-late	DegreeOnly	MEASURED_FULLL10	0.00568 / 253.30	Measured, non-Pareto
LI MEDIUM	Early-to-late	NoEdge	MEASURED_FULLL10	0.00515 / 243.66	Measured, non-Pareto
LI MEDIUM	Early-to-late	RecentWindow	MEASURED_FULLL10	0.00657 / 129.07	Pareto
LI MEDIUM	Early-to-late	SAGE-edge h64	MEASURED_FULLL10	0.01030 / 249.24	Pareto
LI MEDIUM	Early-to-late	ShuffledEdge	MEASURED_FULLL10	0.00551 / 246.03	Measured, non-Pareto
LI MEDIUM	Late-window	DegreeCap	MEASURED_FULLL10	0.00256 / 251.89	Measured, non-Pareto
LI MEDIUM	Late-window	DegreeOnly	MEASURED_FULLL10	0.00778 / 303.13	Measured, non-Pareto
LI MEDIUM	Late-window	NoEdge	MEASURED_FULLL10	0.00600 / 294.95	Measured, non-Pareto
LI MEDIUM	Late-window	RecentWindow	MEASURED_FULLL10	0.00661 / 156.28	Pareto
LI MEDIUM	Late-window	SAGE-edge h64	MEASURED_FULLL10	0.01105 / 300.67	Pareto
LI MEDIUM	Late-window	ShuffledEdge	MEASURED_FULLL10	0.00634 / 294.61	Measured, non-Pareto

Scope and reading. Pareto treatment is computed only within the same variant and protocol. Runtime is runner elapsed time, not a hardware-normalized complexity or energy measure. Blocked rows remain nonnumeric.

RecentWindow reduces aggregated Medium mean runtime from 275.19 to 141.65 seconds, while changing AUPRC by -0.01324 , AUROC by $+0.02525$, and F1 by $+0.00419$. Pareto flags are computed only within the same variant/protocol cell.

TABLE XXIX
IBM BASELINE GRID: AUPRC AND F1@0.5 WINNERS AND RANK AGREEMENT.

Variant	Protocol	Feasible n	AUPRC winner	F1 winner	ρ	Disagree
HI MEDIUM	Early-to-late	3	HistGB	HistGB	1.000	No
HI MEDIUM	Late-window	3	HistGB	LogReg	-1.000	Yes
HI SMALL	Early-to-late	3	HistGB	HistGB	1.000	No
HI SMALL	Late-window	3	HistGB	HistGB	1.000	No
LI MEDIUM	Early-to-late	3	HistGB	SAGE-edge h32	-0.500	Yes
LI MEDIUM	Late-window	3	HistGB	SAGE-edge h32	-0.500	Yes
LI SMALL	Early-to-late	3	HistGB	HistGB	1.000	No
LI SMALL	Late-window	3	HistGB	HistGB	1.000	No

Scope and reading. Spearman correlation is computed over the exact feasible set in each row. F1 uses the saved threshold of 0.5; disagreement is not asserted to be threshold-invariant.

TABLE XXX
IBM GRAPH GRID: AUPRC AND F1@0.5 WINNERS AND RANK AGREEMENT.

Variant	Protocol	Feasible n	AUPRC winner	F1 winner	ρ	Disagree
HI MEDIUM	Early-to-late	6	SAGE-edge h64	DegreeCap	-0.486	Yes
HI MEDIUM	Late-window	6	SAGE-edge h64	DegreeCap	-0.771	Yes
HI SMALL	Early-to-late	7	GINE h64	DegreeCap	0.857	Yes
HI SMALL	Late-window	7	GINE h64	DegreeCap	0.786	Yes
LI MEDIUM	Early-to-late	6	SAGE-edge h64	DegreeCap	-0.086	Yes
LI MEDIUM	Late-window	6	SAGE-edge h64	DegreeCap	-0.029	Yes
LI SMALL	Early-to-late	7	GINE h64	DegreeCap	0.607	Yes
LI SMALL	Late-window	7	GINE h64	DegreeCap	0.571	Yes

Scope and reading. Spearman correlation is computed over the exact feasible set in each row. F1 uses the saved threshold of 0.5; disagreement is not asserted to be threshold-invariant.

Runtime is not normalized across hardware or implementations.

a) *Sensitivity location.*: The complete context rows, including reference/candidate means, intervals, effect sizes, descriptive p-values, and source paths, remain at `results/tkde_rebuild/IBM_MATCHED_ABLATION_CONTEXT_EFFECTS.csv`. Their checksum and schema are reported in Table XLIII.

XIV. RANK-VERSUS-DECISION ANALYSES

A. Baseline-grid disagreement

Table XXIX compares AUPRC and fixed-threshold F1 winners in the eight baseline contexts. The feasible set contains the same three configurations in every row.

The AUPRC and F1 winners differ in 3 of 8 baseline contexts. This is not a contradiction: AUPRC summarizes the score ordering, while F1 evaluates one threshold. Rank correlation is shown to expose the magnitude of agreement rather than only a categorical winner change.

B. Graph-grid disagreement

Table XXX performs the same comparison in the exact feasible graph-construction set.

The winners differ in all 8 graph-grid contexts, giving 11 of 16 disagreements across the two families. For example, in HI-Medium late-window, SAGE-edge h64 ranks first by AUPRC but sixth by F1; DegreeCap ranks sixth by AUPRC but first by F1. Medium GINE is not in the feasible set, and the sender-receiver identity row is not counted as an independent method.

a) *Interpretive limit.*: The count 11 of 16 is specific to threshold 0.5. It does not establish disagreement under every threshold or review budget, and higher AUROC does not automatically imply better operational decisions.

XV. REVIEW-BUDGET AND CALIBRATION DIAGNOSTICS

A. One-percent review capacity

Review-budget metrics ask how much fraud is found when only the top $b\%$ of scores can be reviewed. Table XXXI reports the central 1% operating point for five saved-prediction policies on each dataset.

On DGraphFin, GraphSafe conservative averages precision 0.0229 and recall 0.0209, compared with 0.0156 and 0.0143 for simple averaging. On Elliptic, the direction reverses: GraphSafe averages precision 0.7624 and recall 0.1176, while simple averaging reaches 0.8274 and 0.1276. The complete 0.5%, 1%, and 2% rows remain in `results/tkde_rebuild/REVIEW_BUDGET_CURVES.csv`.

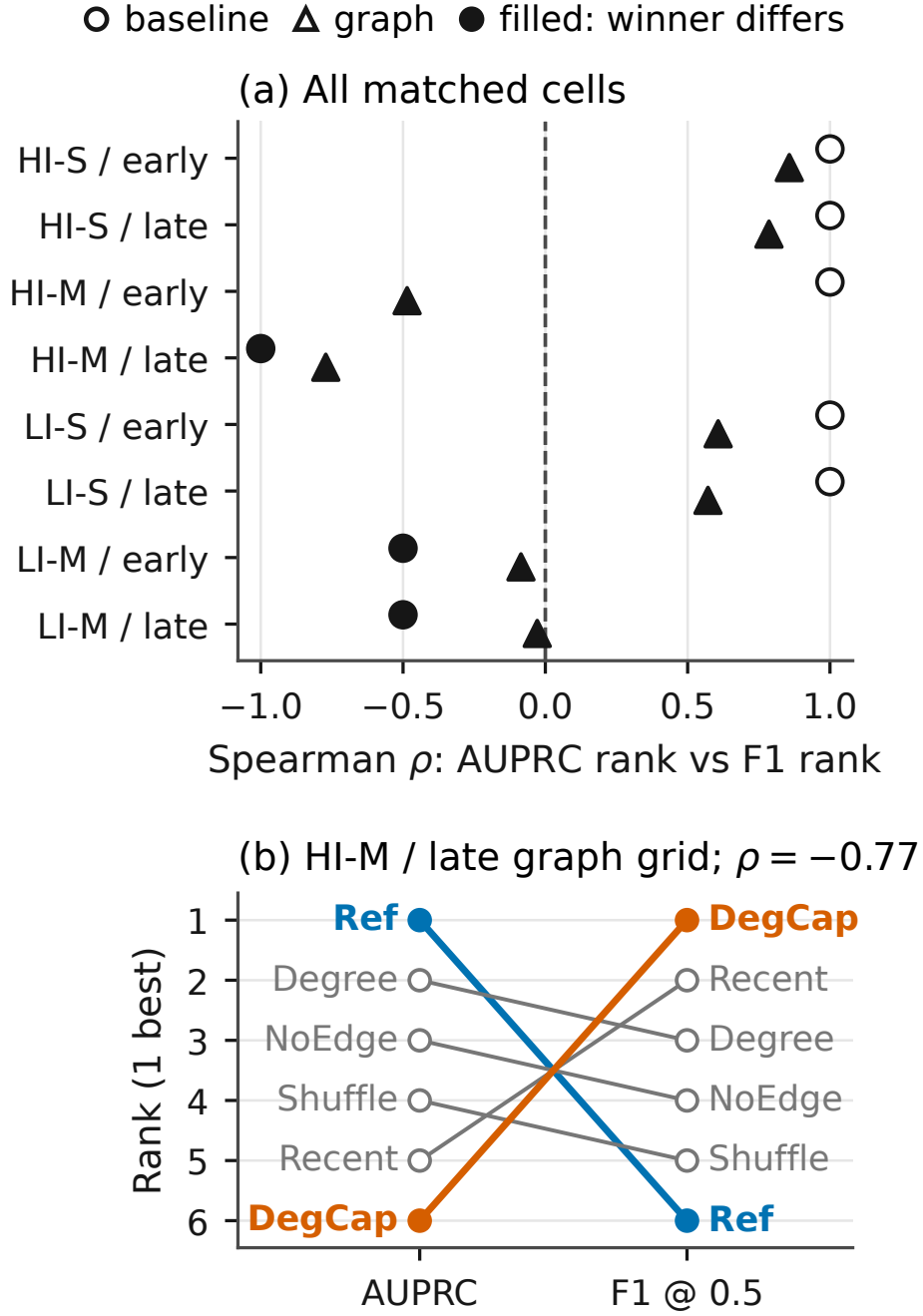


Fig. 5. AUPRC and F1@0.5 rank divergence within exact IBM feasibility sets. The plot makes the operating-point dependence visible without pooling scales or blocked configurations.

TABLE XXXI
PRECISION AND RECALL AT A 1% REVIEW BUDGET.

Dataset	Policy	n	Precision	SD	Recall (SD)
DGraphFin	Best validation branch	10	0.0150	0.0058	0.0138 (0.0053)
DGraphFin	Feature branch	10	0.0161	0.0124	0.0147 (0.0114)
DGraphFin	Graph branch	10	0.0150	0.0058	0.0138 (0.0053)
DGraphFin	GraphSafe conservative	10	0.0229	0.0062	0.0209 (0.0056)
DGraphFin	Simple average	10	0.0156	0.0078	0.0143 (0.0071)
Elliptic	Best validation branch	10	0.7785	0.0482	0.1201 (0.0074)
Elliptic	Feature branch	10	0.6481	0.0635	0.0999 (0.0098)
Elliptic	Graph branch	10	0.8353	0.0359	0.1288 (0.0055)
Elliptic	GraphSafe conservative	10	0.7624	0.0432	0.1176 (0.0067)
Elliptic	Simple average	10	0.8276	0.0556	0.1276 (0.0086)

Scope and reading. The six model/protocol contexts are averaged within seed before the ten-seed summary. Review-budget results are capacity-conditioned and do not replace AUPRC.

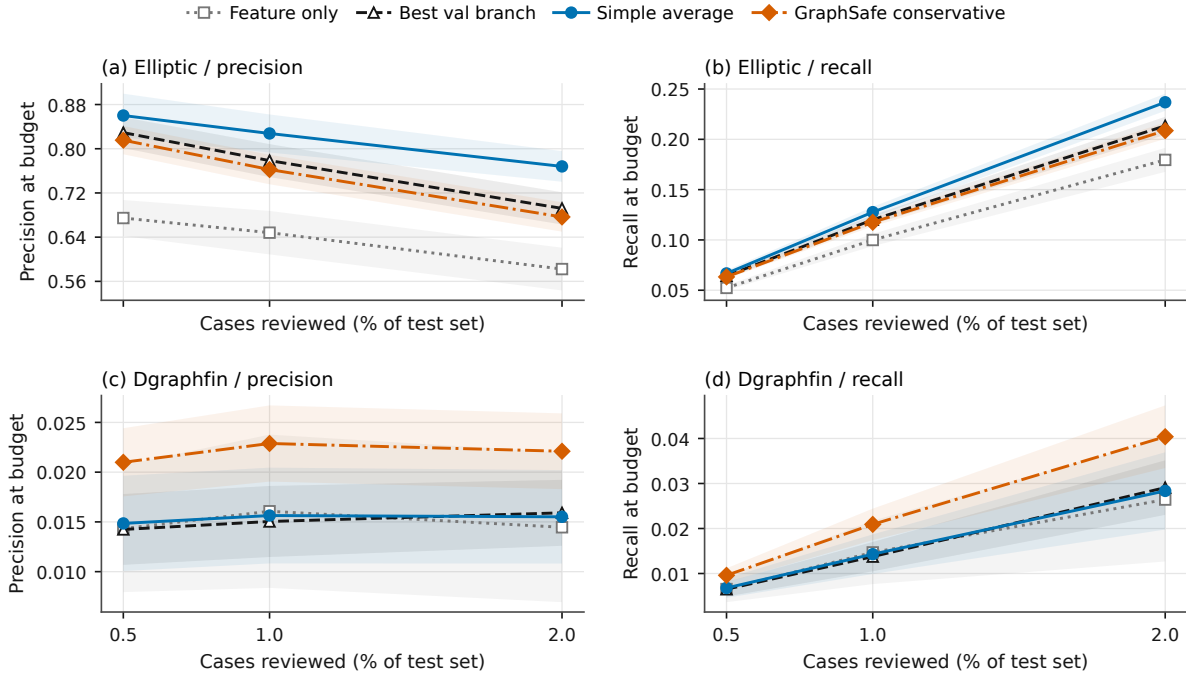


Fig. 6. Precision and recall across the three declared review capacities. Uncertainty uses ten seed blocks after averaging six fixed contexts within seed.

B. Calibration

Table XXXII reports ECE and Brier score from saved predictions.

On DGraphFin, GraphSafe’s mean ECE is 0.2751 and Brier score is 0.09094, compared with 0.2826 and 0.09295 for simple averaging. On Elliptic, simple averaging is better on both diagnostics (ECE 0.09631 versus 0.09820; Brier 0.07059 versus 0.07899).

a) Limit. These are fixed-dataset diagnostics, not guarantees of calibration in a bank population. Capacity, delayed labels, investigator actions, and changing prevalence can alter the operating surface.

XVI. GRAPHSAFE-TTA CASE STUDY

A. Policy definition

GraphSafe is a post-training switch over aligned feature and graph prediction files. It does not retrain either branch. For each prediction it combines absolute score disagreement, within-split percentile-rank disagreement, and mean binary entropy. Validation-set 95th percentiles scale the terms:

$$\rho_i = \frac{d_i^{\text{score}}/a_s + d_i^{\text{rank}}/a_r + 0.5h_i/a_h}{2.5}. \quad (12)$$

TABLE XXXII
SAVED-PREDICTION CALIBRATION DIAGNOSTICS.

Dataset	Policy	n	ECE	SD	Brier (SD)
DGraphFin	Best validation branch	10	0.28443	0.02093	0.09632 (0.01160)
DGraphFin	Feature branch	10	0.28076	0.02877	0.09356 (0.01546)
DGraphFin	Graph branch	10	0.28443	0.02093	0.09632 (0.01160)
DGraphFin	GraphSafe conservative	10	0.27507	0.02032	0.09094 (0.01054)
DGraphFin	Simple average	10	0.28260	0.02234	0.09295 (0.01241)
Elliptic	Best validation branch	10	0.08760	0.01392	0.07341 (0.00807)
Elliptic	Feature branch	10	0.13074	0.00643	0.09912 (0.00441)
Elliptic	Graph branch	10	0.07298	0.00779	0.06515 (0.00580)
Elliptic	GraphSafe conservative	10	0.09820	0.01727	0.07899 (0.00848)
Elliptic	Simple average	10	0.09631	0.00381	0.07059 (0.00255)

Scope and reading. ECE uses ten bins and is descriptive. These fixed-dataset summaries do not establish calibrated probabilities in a deployment population.

A validation search chooses cutoff γ and decision threshold τ , and

$$q_i^{\text{GS}} = \begin{cases} q_i^{\text{graph}}, & \rho_i \leq \gamma, \\ q_i^{\text{feature}}, & \rho_i > \gamma. \end{cases} \quad (13)$$

Test labels are used only for final evaluation. Comparators include each branch, the validation-selected branch, and simple averaging. Additional saved-output diagnostics use validation-weighted averaging, temperature scaling, prior correction, and rolling thresholds [17], [21], [22].

B. Aggregate policy outcomes

Table XXXIII first averages six graph-model/protocol contexts within seed and then summarizes ten seeds.

TABLE XXXIII: GraphSafe and comparator outcomes after seed-block aggregation.

Dataset	Policy	n	F1	AUPRC	Recall@1%	Cost risk
DGraphFin	Best validation branch	10	0.03624	0.01641	0.01375	0.47266
DGraphFin	Feature branch	10	0.03241	0.01390	0.01469	0.59243
DGraphFin	Graph branch	10	0.03624	0.01641	0.01375	0.47266
DGraphFin	GraphSafe conservative	10	0.04257	0.01901	0.02092	0.29128
DGraphFin	Simple average	10	0.03647	0.01625	0.01430	0.45220
Elliptic	Best validation branch	10	0.53247	0.49281	0.12005	0.17972
Elliptic	Feature branch	10	0.51700	0.43718	0.09994	0.17929
Elliptic	Graph branch	10	0.53252	0.51525	0.12881	0.18203
Elliptic	GraphSafe conservative	10	0.53325	0.48762	0.11756	0.17831
Elliptic	Simple average	10	0.55130	0.53292	0.12762	0.17086

Scope and reading. Each row averages six fixed contexts within seed, then summarizes ten seeds. The 1:5 cost is illustrative. DGraphFin descriptive gains and Elliptic negative comparisons are both retained.

On DGraphFin, GraphSafe conservative has mean F1 0.04257, AUPRC 0.01901, Recall@1% 0.02092, and cost risk 0.29128; simple averaging has 0.03647, 0.01625, 0.01430, and 0.45220. On Elliptic, simple averaging is stronger on F1 (0.55130 versus 0.53325), AUPRC (0.53292 versus 0.48762), Recall@1% (0.12762 versus 0.11756), and cost risk (0.17086 versus 0.17831). Both the favorable and negative comparisons are central to the bounded interpretation.

C. Multiplicity-aware comparison

Table XXXIV reports the 12 GraphSafe-conservative contrasts against simple averaging. The full correction family contains 48 comparator-metric tests.

TABLE XXXIV: GraphSafe conservative minus simple-average paired effects.

Dataset	Metric	Direction	Δ	95% CI	Holm p
DGraphFin	auprc	higher	+0.00275	[0.00113, 0.00451]	0.3906
DGraphFin	average block regret	lower	+0.16112	[0.06278, 0.28531]	0.2246
DGraphFin	cost sensitive risk	lower	+0.16092	[0.06409, 0.28513]	0.2246
DGraphFin	f1	higher	+0.00610	[0.00370, 0.00868]	0.1094
DGraphFin	recall at 1pct	higher	+0.00662	[0.00402, 0.00944]	0.0938

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TABLE XXXIV: GraphSafe conservative minus simple-average paired effects. (continued)

Dataset	Metric	Direction	Δ	95% CI	Holm p
DGraphFin	worst block regret	lower	+0.16603	[0.06420, 0.29419]	0.2246
Elliptic	auprc	higher	-0.04530	[-0.05466, -0.03622]	0.0938
Elliptic	average block regret	lower	-0.00667	[-0.00825, -0.00515]	0.0938
Elliptic	cost sensitive risk	lower	-0.00745	[-0.00887, -0.00610]	0.0938
Elliptic	f1	higher	-0.01804	[-0.02384, -0.01223]	0.0938
Elliptic	recall at 1pct	higher	-0.01006	[-0.01371, -0.00648]	0.1094
Elliptic	worst block regret	lower	-0.03058	[-0.04066, -0.01922]	0.1094

Scope and reading. The correction family contains all 48 saved-prediction comparator tests, not only the 12 rows shown here. No adjusted p-value is below 0.05; therefore this case does not establish universal dominance or ranking repair.

The DGraphFin GraphSafe AUPRC difference is 0.00275 with 95% CI [0.00113, 0.00451], but Holm $p = 0.3906$. The Elliptic difference is -0.04530 , CI $[-0.05466, -0.03622]$, with Holm $p = 0.0938$. No one of the 48 adjusted tests is below 0.05.

a) *What this case establishes.*: The saved-output surface demonstrates how a typed support system retains mixed operational evidence and strong comparators. It does not establish universal GraphSafe dominance, ranking repair, automatic AML deployment, or a correction of rank metrics by calibration.

XVII. RESOURCE-BOUNDARY CASE STUDIES

The resource coordinate records what was attempted under which envelope and how the benchmark treats the outcome. Table XXXV is the compact index; the five cases below give the reviewer-facing interpretation.

TABLE XXXV: Measured resource outcomes and benchmark treatment.

Evaluation cell	Declared envelope / observed event	Outputs	Status	Benchmark treatment
IBM AML HI-Large	safe resource guard	0 results; 0 predictions	SAFE_RESOURCE_BLOCKED	not in predictive ranking
IBM AML LI-Large	safe resource guard	0 results; 0 predictions	SAFE_RESOURCE_BLOCKED	not in predictive ranking
IBM AML HI-Medium GINE h64	single Tesla T4 CUDA OOM	0 of 20 planned results; 0 predictions	RESOURCE_BLOCKED_T4_CUDA_OOM	not in predictive ranking
IBM AML LI-Medium GINE h64	single Tesla T4 CUDA OOM	0 of 20 planned results; 0 predictions	RESOURCE_BLOCKED_T4_CUDA_OOM	not in predictive ranking
DGraphFin GAT h64/l2	Tesla T4 CUDA OOM	0 of 20 planned results; 0 predictions	BLOCKED_T4_OOM	memory-reduced h32/l1 diagnostic is not a replacement
DGraphFin GraphSAGE pool rerun	max- larger GPU required	0 results; 0 predictions	BLOCKED_WAITING_FOR_GPU	no imported result CSV

Scope and reading. A blocked cell is unmeasured and excluded from predictive ranks, matched means, and Pareto sets. A reduced diagnostic cannot replace a fixed blocked configuration.

A. IBM AML Large guards

HI-Large and LI-Large stopped at the safe resource guard and produced zero result and prediction files. Evidence exists for the guard and planned configuration; predictive evidence does not exist. The safe wording is that Large lies outside the admitted resource envelope and is unmeasured. It is not permissible to rank Large, infer underperformance, or claim full-10 coverage.

B. Medium GINE T4 out-of-memory

HI-Medium and LI-Medium GINE h64 each encountered single-T4 CUDA OOM and produced zero of 20 planned outputs. Small GINE performance and the Medium failure record both exist. Medium GINE performance does not. It is excluded from Medium winner sets, matched means, and Pareto sets rather than imputed.

C. Fixed DGraphFin GAT T4 out-of-memory

The fixed h64/l2 GAT lane has zero of 20 planned results and predictions under the T4 envelope. A memory-reduced h32/l1 diagnostic changes architecture and cannot substitute for the blocked fixed cell. The benchmark may compare their feasibility statuses, not their predictive scores.

D. RB18 and larger-GPU extension status

The DGraphFin GraphSAGE max-pool extension has preparation commands and status files but no imported result CSV. A larger GPU was requested; no validated runtime record establishes P100, A100, or H100 performance. The safe wording is waiting-for-GPU/unmeasured, not hardware-backed evidence.

E. Memory-reduced exploratory architectures

Memory-reduced variants can diagnose whether a code path runs under a smaller envelope. They remain exploratory because width, depth, sampling, or features differ from the fixed target. Directory presence, notebook generation, or partial predictions cannot fill a claim scoped to the original architecture.

F. False-promotion protections

Table XXXVI explains how count-only logic could incorrectly promote failed imports, duplicate labels, aliases, diagnostics, and zero-output plans.

TABLE XXXVI: False-promotion patterns guarded by the artifact layer.

Audit	Family	Result risk	Prediction risk	Correct treatment	Why count-only logic fails
FP01	V22 failed/noncomparable imports	38	38	EXCLUDED_INTEGRITY	missing seed, failed return-code, duplicate/conflicting merge metadata
FP02	V24 RB41 duplicate stress labels	240	240	DEDUPLICATE_TO_120_BASE_CELLS	stress label is never passed to benchmark harness; all performance metrics repeat
FP03	V24 memory-reduced DGraphFin GAT h32/I1	20	20	DIAGNOSTIC_ONLY	different width/depth; canonical lock says not comparable
FP04	V26 legacy P100-named lane	12	12	NORMALIZE_ALIAS_TO_T4/CUDA0	runtime records identify cuda:0 and validation normalizes the alias; path names are not hardware evidence
FP05	IBM AML Large status/runbook artifacts	0	0	RESOURCE_BLOCKED	canonical lock contains zero results and predictions
FP06	Medium GINE blocked plans	0	0	RESOURCE_BLOCKED	two T4-OOM rows; zero results and predictions
FP07	RB18 larger-GPU preparation	0	0	BLOCKED_WAITING_FOR_GPU	validation reports no imported result CSV

Scope and reading. The 310 result and 310 prediction files at risk are not additional evidence. Zero-output plans remain visible because directories and notebooks can otherwise be mistaken for completed runs.

Across metric-bearing artifacts, 310 result and 310 prediction files are at risk of naive promotion. The count includes 240 duplicate V24 labels, 38 failed/noncomparable V22 imports, 20 memory-reduced GAT diagnostics, and 12 legacy hardware-alias files. These remain provenance or validation cases, not additional performance evidence.

a) *General treatment.*: Resource-blocked cells are unmeasured and predictively unordered. A future run on different hardware could produce evidence, but no direction follows from that possibility.

XVIII. CLAIM LEDGER AND PROVENANCE MAP

The frozen ledger contains 22 typed claims: 12 supported, three diagnostic-only, two refuted-in-scope, two resource-blocked, two supported theoretically, and one supported with a resource boundary. The following five tables preserve every claim ID, exact scope string, status, permitted wording, and prohibited generalization exactly once. Full matched evidence-unit identifiers remain in the machine-readable ledger.

A. Protocol and architecture claims

These claims distinguish an observed Elliptic top-rank reversal and architecture-dependent visibility effects from the prohibited universal graph-harm narrative. C19 further prevents results from contracts with different claim-relevant coordinates from being treated as interchangeable.

B. IBM baseline, construction, and scale claims

The IBM group separates measured Small/Medium effects, the Small-only GINE result, Large and Medium-GINE resource outcomes, and the sender-receiver identity diagnostic. The quantifier boundary matters because a feasible Small winner cannot be promoted to an all-scale claim.

TABLE XXXVII
PROTOCOL AND ARCHITECTURE CLAIMS.

ID	Status	Exact scoped subject	Permitted wording	Prohibited generalization
C01	SUPPORTED	Elliptic; MLP/GCN/GraphSAGE; strict versus isolated; seeds 1-10	the evaluated Elliptic leaderboard reverses at the top under the two visibility contracts	evaluation protocols always reverse rankings
C02	SUPPORTED	Elliptic and DGraphFin; MLP; strict versus isolated; seeds 1-10	the MLP negative control is unchanged by the graph-only visibility intervention	the two protocols are identical
C03	SUPPORTED	Elliptic and DGraphFin; GCN and GraphSAGE; strict versus isolated; seeds 1-10	effects depend on architecture and dataset in the evaluated grid	graph isolation is always harmful or always beneficial
C04	REFUTED_IN_SCOPE	all datasets, constructions, protocols, and graph models	the evaluated evidence rejects a uniform graph-harm narrative	GNNs fail universally; graph methods are always harmful
C19	SUPPORTED_ THEORETICALLY	typed deployment contracts	protocol results are contract-relative	any numerical difference is caused solely by graph visibility

Scope and reading. Status and scope text are copied from the frozen 22-claim ledger. Exact matched evidence-unit identifiers remain in the machine-readable ledger rather than in these reviewer-facing cells.

TABLE XXXVIII
IBM AML BASELINE, CONSTRUCTION, AND SCALE CLAIMS.

ID	Status	Exact scoped subject	Permitted wording	Prohibited generalization
C05	SUPPORTED	IBM AML HI/LI Small/Medium; two temporal protocols; HistGB/LogReg/GraphSAGE-h32; seeds 1-10	HistGB leads mean AUPRC in all eight evaluated baseline-grid contexts	trees are universally best for AML
C07	SUPPORTED	HI/LI Small/Medium; both protocols; edge-aware GraphSAGE h64 versus NoEdge/ShuffledEdge; seeds 1-10	the original edge-feature representation outperforms the two matched controls on mean AUPRC in this grid	edge features causally improve all fraud models
C08	SUPPORTED	IBM AML HI/LI Small/Medium; both protocols; DegreeCap versus edge-aware GraphSAGE reference	degree capping exposes metric- and regime-dependent tradeoffs	degree capping improves the benchmark
C09	SUPPORTED	IBM AML HI/LI Medium; both protocols; complete V27/V28 graph configurations	RecentWindow lies on a metric-dependent speed/performance tradeoff in the Medium grid	RecentWindow is Pareto-optimal for every objective
C10	SUPPORTED_WITH_ RESOURCE_BOUNDARY	IBM AML HI/LI Small for performance; HI/LI Medium for resource status	GINE leads mean AUPRC on the feasible Small cells; Medium remains resource-blocked	GINE is best across Small and Medium; Medium GINE performs poorly
C11	RESOURCE_BLOCKED	IBM AML HI-Large and LI-Large	Large lies outside the admitted resource envelope and is unmeasured	Large performs worse; Large full10 evidence
C12	RESOURCE_BLOCKED	IBM AML HI-Medium and LI-Medium; GINE h64	Medium GINE is outside the measured performance set	Medium GINE underperforms
C13	DIAGNOSTIC_ONLY	IBM AML V28 sender-receiver configuration versus V27 reference	contract restatement/equivalence check	a distinct sender-receiver method independently replicated the result

Scope and reading. Status and scope text are copied from the frozen 22-claim ledger. Exact matched evidence-unit identifiers remain in the machine-readable ledger rather than in these reviewer-facing cells.

TABLE XXXIX
RANK, DECISION, PREVALENCE, AND CALIBRATION CLAIMS.

ID	Status	Exact scoped subject	Permitted wording	Prohibited generalization
C06	SUPPORTED	IBM AML Small/Medium locked V26-V28 cells	rank and decision metrics answer different questions and disagree in the evaluated cells	higher AUROC implies better operational decisions
C14	DIAGNOSTIC_ONLY	IBM AML V26-V28 cells with recorded test prevalence	prevalence-normalized AUPRC contextualizes raw values	normalized AUPRC is prevalence-invariant performance
C18	SUPPORTED_ THEORETICALLY	binary scoring with no tie-order changes	rank invariance under strictly monotone transforms	calibration necessarily improves AUPRC/AUROC

Scope and reading. Status and scope text are copied from the frozen 22-claim ledger. Exact matched evidence-unit identifiers remain in the machine-readable ledger rather than in these reviewer-facing cells.

TABLE XL
GRAPHSAFE-TTA CLAIMS.

ID	Status	Exact scoped subject	Permitted wording	Prohibited generalization
C15	DIAGNOSTIC_ONLY	DGraphFin saved RB15/RB15b policy rows and declared cost setting	bounded DGraphFin decision-risk case study	GraphSafe universally dominates; GraphSafe repairs ranking
C16	SUPPORTED	Elliptic saved RB15/RB15b policy rows	GraphSafe does not dominate simple averaging on Elliptic	GraphSafe improves all datasets
C17	REFUTED_IN_SCOPE	all datasets/protocols/models	GraphSafe is a bounded decision-analysis case study with mixed outcomes	universal dominance; ranking repair

Scope and reading. Status and scope text are copied from the frozen 22-claim ledger. Exact matched evidence-unit identifiers remain in the machine-readable ledger rather than in these reviewer-facing cells.

TABLE XLI
EVIDENCE-INTEGRITY AND RESOURCE CLAIMS.

ID	Status	Exact scoped subject	Permitted wording	Prohibited generalization
C20	SUPPORTED	all benchmark cells	unmeasured under the declared resource envelope	blocked model underperforms
C21	SUPPORTED	V26, V27, V28 canonical imported locks	840/840 locked IBM AML result/prediction files	840 independent configurations; blocked outputs included
C22	SUPPORTED	sandbox mutations of the generated evidence inventory	the validator rejects the tested incomplete/invalid cases	the gate proves all scientific claims true

Scope and reading. Status and scope text are copied from the frozen 22-claim ledger. Exact matched evidence-unit identifiers remain in the machine-readable ledger rather than in these reviewer-facing cells.

C. Rank, decision, prevalence, and calibration claims

These claims keep rank metrics, fixed-threshold decisions, prevalence-normalized diagnostics, and monotone calibration distinct. They rule out language that treats one operating point or normalized lift as a universal performance measure.

D. GraphSafe-TTA claims

The GraphSafe group retains DGraphFin descriptive benefits and Elliptic negative comparisons. The universal-improvement claim is refuted in scope, so the case remains secondary and decision-level.

E. Evidence integrity and resources

C20 makes blocked cells predictively unordered; C21 records the 840/840 IBM artifact count without calling the files independent configurations; C22 limits the validator result to the tested failure modes.

F. From printed claim to canonical lock

Table XLII gives the short reviewer path from each core scientific family to its canonical evidence lock.

TABLE XLII: Reviewer-facing experiment-family to evidence-lock map.

Scientific family	Eligibility	Cells	Datasets	Canonical lock
IBM AML baseline grid	main-paper eligible	24	IBM AML-Data	results/v26_imported/V26_IMPORTED_EVIDENCE_LOCK.json
IBM AML construction/feature ablation	main-paper eligible	52	IBM AML-Data	results/v28_imported/V28_ALL_RUNS_EVIDENCE_LOCK.json
IBM AML stronger graph study	main-paper eligible	8	IBM AML-Data	results/v27_imported/V27_STRONGER_GRAPH_EVIDENCE_LOCK.json
core strict/isolated/transductive benchmark	main-paper eligible	18	dgraphfin;elliptic	results/runs_rb09v3/ARTIFACT_FAMILY.json
loss/control/architecture extension	supplement-only	64	dgraphfin;elliptic	kaggle_workspace/manifests/V22_FINAL_GPU_EVIDENCE_LOCK.json
review-budget/worst-block/cost sensitivity	supplement-only	1	Elliptic; DGraphFin	results/runs_rb17_review_budget_worst_block/import_manifest.json
temporal-shift/perturbation stress	supplement-only	48	dgraphfin;elliptic	results/v24_imported/V24_IMPORTED_EVIDENCE_LOCK.json

Continued on next page

TABLE XLII: Reviewer-facing experiment-family to evidence-lock map. (continued)

Scientific family	Eligibility	Cells	Datasets	Canonical lock
validation-selected conservative policy	GraphSafe main case-study eligible	1	Elliptic; DGraphFin	results/runs_rb15b_graphsafe_conservative_policy/import_manifest.json

Scope and reading. The complete 247-record evidence inventory and 6,796-record scalar map remain machine-readable. This table shows only the core paths a reviewer needs to follow the printed results.

The machine-readable evidence inventory contains 247 records, and the scalar provenance map contains 6,796 records. Table XLII identifies their canonical locations. The compact table is a navigation layer, not a replacement for either exhaustive map.

XIX. REPRODUCTION AND ARTIFACT INDEX

A. Deterministic no-training path

The visual rebuild consumes the frozen scientific surfaces and regenerates only reviewer-facing tables. From the repository root:

```
gmn_env/bin/python \
  scripts/tkde_visual_rebuild/build_curated_supplement_tables.py
```

The generator verifies recorded frozen hashes where available, asserts the 22 claim IDs and status counts, checks the four exhaustive row counts and hashes, and rejects active portrait-table fragments containing tiny, scriptsize, resizebox, landscape wrappers, or forced page commands.

The supplement bibliography cycle is:

```
cd paper_tkde/supplement
pdflatex -interaction=nonstopmode -halt-on-error supplement.tex
bibtex supplement
pdflatex -interaction=nonstopmode -halt-on-error supplement.tex
pdflatex -interaction=nonstopmode -halt-on-error supplement.tex
```

This path regenerates the visual representation from already derived, locked evidence. The broader saved-evidence analysis path remains documented in the baseline command log. Neither path is a claim that full training was rerun.

B. Exhaustive rows moved to the artifact

Table XLIII records the full machine-readable destination, row count, path, SHA-256, and schema prefix for each raw table removed from the PDF.

TABLE XLIII: Machine-readable destination of the four exhaustive tables removed from the PDF.

Evidence family	Rows	Artifact path	SHA-256	Schema prefix
RB09 SEED GRID	180	results/runs_rb09v3/runs.csv	d9f77bfd14ccaf157780858e35ecfee96af1e2fbf60f3559ad888d14c0ace2e9	dataset, protocol, split_name, model, seed, early_stopping_metric; ...
V22 PAIRED TESTS	198	manuscript_assets/tables/V22_STAT_TESTS_FULLL10.csv	513f1fb5c2dbae2f63255fd384c62fe86394a15746c31792a92a1f287b8ddadd	family, dataset, protocol, fixed_model, comparison_axis, left; ...
IBM SEED GRID	840	results/tkde_rebuild/IBM_IMPORTED_SEED_ROWS.csv	dae5d51178b9a746946c9fc05bf31e74840b6663d69031597ef2be45d5e766b6	version, variant, size, regime, protocol, config; ...
IBM CONTEXT EFFECTS	208	results/tkde_rebuild/IBM_MATCHED_ABLATION_CONTEXT_EFFECTS.csv	626f0e014a0b8d339fe344bb736d513ed63a6ec435be6104acad1f53f7bbe202	config, size, metric, variant, protocol, n_seed_pairs; ...

Scope and reading. Full ordered schemas, complete hashes, and the common regeneration command are recorded in results/tkde_visual_rebuild/RAW_TABLE_ARTIFACT_INDEX.csv and results/tkde_visual_rebuild/RAW_TABLE_ARTIFACT_INDEX.md. No row is deleted or converted to a PDF-only value.

The complete ordered schemas and commands live in results/tkde_visual_rebuild/RAW_TABLE_ARTIFACT_INDEX.csv and RAW_TABLE_ARTIFACT_INDEX.md. The four row counts are exactly 180, 198, 840, and 208. Their source SHA-256 values are checked before any curated fragment is written.

C. Representation and completeness

The 180 node seeds become three matched metric tables with strict/isolated means, paired deltas, intervals, effect sizes, adjusted tests, and n . The 198 legacy tests become a family/metric composition table while preserving every original test row in CSV. The 840 IBM seeds become baseline, matched-effect, rank, runtime, and feasibility aggregates. The 208 context sensitivities remain available for heterogeneity inspection while the PDF reports the dependence-aware ten-seed inference.

No missing value is converted to zero, no blocked configuration receives a numeric cell, and no unmatched feasibility set is averaged. The artifact retains full result paths and source paths that would be distracting and fragile in typeset pages.

D. Provenance and packaging boundary

Canonical imports and locks are read-only. Generated LaTeX fragments live under `paper_tkde/supplement/tables`; the curated-table manifest records their hashes and portrait/footnotesize invariants. A clean release should include the four machine-readable sources, their index, the claim/evidence ledger, scalar map, lock paths, checksums, and excluded-file manifest while omitting raw restricted data, credentials, private absolute paths, caches, macOS metadata, predictions that cannot be redistributed, and build debris.

a) *Reproducibility limit.*: Deterministic reconstruction from saved evidence is verified. Fresh data acquisition, model training, GPU execution, and an external artifact badge are separate activities and are not claimed here.

XX. EXTENDED LIMITATIONS AND ETHICS

A. Dataset and task scope

The evidence covers two public real graphs and four measured regimes from one synthetic IBM generator. Prediction units differ across transaction nodes, account nodes, and transaction edges. No result estimates performance for an unobserved institution, jurisdiction, payment network, fraud typology, or laundering adaptation. DGraphFin time is an incident-edge-derived bucket construction, Elliptic uses fixed dataset steps, and IBM HI/LI are synthetic regimes rather than bank populations.

B. Architecture and construct scope

The primary real-data grid contains MLP, GCN, and GraphSAGE rather than a broad temporal, heterogeneous, or transformer survey. Strict-to-isolated changes graph access and may change several structural properties at once; it shows sensitivity, not a unique mechanism. V24 stress labels are construct-invalid as separate conditions, and the sender-receiver row is an identity alias. The IBM early-to-late classifier uses first-50% labels with first-60% label-free histories, so it is not a pure first-50%-information simulation.

C. Statistical scope

Ten seeds quantify optimization variation on fixed datasets and splits, not uncertainty across institutions or future periods. Percentile intervals and Wilcoxon tests inherit that limit. Holm controls the declared analysis families, not every historical exploration. The IBM construction analysis averages four fixed contexts within seed; the 208 context rows remain available because dependence-aware aggregation does not erase heterogeneity. GraphSafe averages six contexts within seed and therefore gives a valid ten-block comparison at the cost of hiding some context variation in its compact summary.

D. Compute and missing cells

IBM Large, Medium GINE, fixed DGraphFin GAT h64/l2, and the max-pool extension remain unmeasured. Another implementation or larger GPU could make them feasible, but no predictive direction follows. Runtime is runner elapsed time within the recorded environment, not a general energy, latency, or hardware-efficiency benchmark.

E. Artifact and validator scope

The rebuild verifies saved-output reconstruction, not clean-machine retraining. The support schema and 14 mutations were designed inside this project and establish implementation conformance only. External curators, independent implementations, and inter-rater studies are absent. The artifact must remain synchronized with the final PDFs, but checksums cannot prove construct validity or scientific truth.

F. Operational and fairness limits

The fixed 1:5 cost, review budgets, and calibration bins are scenarios. Real investigator capacity, delayed labels, investigation quality, interventions, and feedback loops can change value [23]. The datasets do not provide an appropriate surface for a complete fairness or disparate-impact evaluation. No claim is made about case resolution, investigator productivity, monetary loss reduction, regulatory compliance, or equitable deployment.

G. Intended use and harm

Fraud scores can burden investigators and customers through false alerts and can affect access to financial services. The benchmark evaluates ranking and review support; it does not authorize automated account closure, punitive action, or identity inference. Deployment would require institution-specific validation, governance, appeal processes, monitoring, security review, and human oversight. Dataset, model, and deployment cards should preserve intended use, exclusions, provenance, and limitations [24]–[26].

H. Remaining human review

Professor/coauthor scientific review, author metadata, venue declarations, biographies if required, and the current TKDE page and supplementary-material policy remain human gates. Publication-design completion does not remove the empirical limitations above.

REFERENCES

- [1] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in Data Mining: Formulation, Detection, and Avoidance,” *ACM Transactions on Knowledge Discovery from Data*, 2012.
- [2] G. C. Cawley and N. L. C. Talbot, “On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation,” *Journal of Machine Learning Research*, 2010. [Online]. Available: <https://jmlr.org/beta/papers/v11/cawley10a.html>
- [3] S. Kapoor and A. Narayanan, “Leakage and the Reproducibility Crisis in Machine-Learning-Based Science,” *Patterns*, 2023.
- [4] K. Gorman and S. Bedrick, “We Need to Talk about Standard Splits,” in *ACL*, 2019.
- [5] A. Z. Jacobs and H. Wallach, “Measurement and Fairness,” in *FACCT*, 2021.
- [6] A. Reuel, A. Hardy, C. Smith, M. Lamparth, M. Hardy, and M. J. Kochenderfer, “BetterBench: Assessing AI Benchmarks, Uncovering Issues, and Establishing Best Practices,” in *NeurIPS Datasets and Benchmarks*, 2024.
- [7] F. Bordes, C. Ross, J. T. Kao, E. Spiliopoulou, and A. Williams, “Eval Factsheets: A Structured Framework for Documenting AI Evaluations,” *arXiv*, 2025, arXiv:2512.04062.
- [8] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiserson, “Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics,” *KDD 2019 Workshop on Anomaly Detection in Finance*; *arXiv*, 2019, arXiv:1908.02591.
- [9] X. Huang, Y. Yang, Y. Wang, C. Wang, Z. Zhang, J. Xu, L. Chen, and M. Vazirgiannis, “DGraph: A Large-Scale Financial Dataset for Graph Anomaly Detection,” in *NeurIPS Datasets and Benchmarks*, 2022.
- [10] E. Altman, J. Blanuša, L. von Niederhäusern, B. Egressy, A. Anghel, and K. Atasu, “Realistic Synthetic Financial Transactions for Anti-Money Laundering Models,” in *NeurIPS Datasets and Benchmarks*, 2023.
- [11] T. N. Kipf and M. Welling, “Semi-Supervised Classification with Graph Convolutional Networks,” in *ICLR*, 2017, arXiv:1609.02907.
- [12] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive Representation Learning on Large Graphs,” in *NeurIPS*, 2017, arXiv:1706.02216.
- [13] J. H. Friedman, “Greedy Function Approximation: A Gradient Boosting Machine,” *The Annals of Statistics*, 2001.
- [14] W. Hu, B. Liu, J. Gomes, M. Zitnik, P. Liang, V. Pande, and J. Leskovec, “Strategies for Pre-training Graph Neural Networks,” in *ICLR*, 2020, arXiv:1905.12265.
- [15] T. Saito and M. Rehmsmeier, “The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets,” *PLOS ONE*, 2015.
- [16] A. C. Bahnsen, D. Aouada, and B. Ottersten, “Example-Dependent Cost-Sensitive Decision Trees,” *Expert Systems with Applications*, 2015.
- [17] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On Calibration of Modern Neural Networks,” in *ICML*, 2017. [Online]. Available: <https://proceedings.mlr.press/v70/guo17a.html>
- [18] B. Efron, “Bootstrap Methods: Another Look at the Jackknife,” *The Annals of Statistics*, 1979.
- [19] S. Holm, “A Simple Sequentially Rejective Multiple Test Procedure,” *Scandinavian Journal of Statistics*, 1979. [Online]. Available: <https://www.jstor.org/stable/4615733>
- [20] Y. Benjamini and Y. Hochberg, “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing,” *Journal of the Royal Statistical Society, Series B*, 1995.
- [21] M. Saerens, P. Latinne, and C. Decaestecker, “Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure,” *Neural Computation*, 2002.
- [22] Y. Geifman and R. El-Yaniv, “Selective Classification for Deep Neural Networks,” in *NeurIPS*, 2017. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html>
- [23] M. A. Siddiqui, A. Fern, T. G. Dietterich, R. Wright, A. Theriault, and D. W. Archer, “Feedback-Guided Anomaly Discovery via Online Optimization,” in *KDD*, 2018.
- [24] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. D. III, and K. Crawford, “Datasheets for Datasets,” *Communications of the ACM*, 2021.
- [25] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, “Model Cards for Model Reporting,” in *FAT**, 2019.
- [26] M. Pushkarna, A. Zaldivar, and O. Kjartansson, “Data Cards: Purposeful and Transparent Dataset Documentation for Responsible AI,” in *FACCT*, 2022.